

ANALYTIC CONSTRUCTION OF RATIONAL CURVES ON FANO MANIFOLDS

YUN-HENG DU, BIN GUO, AND SONG-YAN XIE

ABSTRACT. Inspired by constructions of entire curves in Oka geometry, we construct rational curves on complex Fano manifolds analytically, by alternately deforming a holomorphic disk to reduce its area and enlarging its source. Positive Ricci curvature yields an area-decreasing deformation, while an affine-lift perturbation enlarges the source at a controlled area cost. The two operations change the area and the weighted derivative by amounts we estimate explicitly, and analytic compactness passes from the resulting disks to a holomorphic sphere. The main result produces a sphere that meets a prescribed compact fiber without being contained in it, with an explicit area bound.

CONTENTS

1. Introduction and main results	1
2. Explicit construction of a relative sphere	6
3. Ricci traces and scalar-constrained variations	14
4. Uniform holomorphic realization and disk operations	18
5. Compactness, marked values, and finite-area disks	31
6. Applications	34
Acknowledgments	35
Funding	35
AI use disclosure	35
References	35

1. INTRODUCTION AND MAIN RESULTS

1.1. The analytic construction problem

We construct rational curves on complex Fano manifolds by explicit analytic operations, with control of the intermediate disks. A Fano manifold X is connected, smooth, and compact, with ample anticanonical line bundle K_X^{-1} .

Rational curves are central to Mori's solution [34] of Hartshorne's conjecture on ample tangent bundles. In his monograph *Ample subvarieties of algebraic varieties*, Hartshorne [25] asked whether a smooth complex projective variety with ample tangent bundle must be projective space. Mori first reduces the problem to characteristic $p > 0$, where the Frobenius map is the key tool in his construction of rational curves. He then shows that rational curves also exist on the original complex manifold.

On the analytic side, Siu and Yau [43] used energy-minimizing harmonic spheres in their 1980 proof of the Frankel conjecture. Their argument uses positive holomorphic bisectional curvature, a stronger condition than the Ricci positivity that Yau's prescribed-Ricci theorem provides on every Fano manifold [50]. In 1982, Yau asked whether any two points of a compact Kähler manifold with positive

Date: September 30, 2026.

2020 Mathematics Subject Classification. Primary 14J45, 32Q15; Secondary 14M22, 32Q10, 32Q45.

Key words and phrases. Fano manifold, rational curve, rational connectedness, holomorphic disk, Ricci curvature.

Ricci curvature could be joined by a chain of rational curves [51, p. 44]. Rational connectedness of Fano manifolds was subsequently established by Campana and by Kollár, Miyaoka, and Mori [6, 31]. An analytic construction remained a separate challenge, which Siu later proposed to approach through complex Monge–Ampère equations, without reduction to positive characteristic [42].

Our approach is based on methods for constructing entire curves in Oka geometry, in particular those of Xie [48] (see also [7, 27, 47]). These methods build entire curves by successively modifying holomorphic disks and enlarging their domains. Guo and Xie [22, Section 5] suggested pursuing such constructions on Fano manifolds with a slow-growth condition, with the aim of producing both entire and rational curves analytically. The construction below realizes this idea in the Fano setting.

A nonconstant entire curve of finite area gives a rational curve by removal of the singularity at infinity (Theorem 2.4). We therefore try to construct such an entire curve as a limit of holomorphic disks with radii tending to infinity. Brody’s lemma in complex hyperbolicity [5] suggests normalizing the derivative at the origin and obtaining convergence from derivative bounds on compact subsets. In our setting the total area must also remain uniformly bounded. Positive Ricci curvature provides the area-decreasing deformations used for this purpose. Derivative concentration may prevent direct convergence, so the spheres arising through bubbling are analyzed separately, retaining their required incidence properties.

To carry out these deformations, we need room to vary holomorphic disks in the target. In the Fano setting, the following folklore conjecture in Oka geometry points to such flexibility; see [19, p. 660].

Conjecture A. *Every Fano manifold is Oka.*

Our recent theorem that rationally connected smooth projective manifolds are Oka–1 [12, Corollary 1.8] gives a first step toward the conjecture, since it includes every Fano manifold. The successive approximation and enlargement of source domains used in that work suggest that there is considerable flexibility in deforming holomorphic disks. This viewpoint motivates the alternating deformation and enlargement procedure developed below. Although the Oka–1 theorem assumes rational connectedness, it is used here only as motivation and plays no role in the proof. Known cases of the Oka conjecture include smooth quadric [20] and cubic [29] Fano hypersurfaces. Guo and Xie have recently proved that every smooth quartic hypersurface of dimension at least three in projective space is Oka [23]; see also [49]. For background on Oka and Oka–1 theory, we refer to [18, 1].

The construction is reminiscent of gathering and sealing a baozi: the disks are alternately deformed and enlarged, while the uniform area bound eventually allows the limiting entire curve to be compactified at infinity.

Area and location in the limit require separate treatment, since holomorphic approximation alone supplies no uniform area bound as the source radii tend to infinity, while classical Brody reparametrization moves the source centers [5]: even disks through a fixed point need not yield an entire curve through that point. Duval addressed this loss of location by replacing derivative selection with control of where area accumulates [13]; see also [14]. For a compact set charged by an Ahlfors current, his theorem produces an entire curve whose intersection with that set has positive area. That localization viewpoint also guides the construction below, but in our finite-area setting we retain position through a base constraint and two marked values instead of invoking the Ahlfors-current theorem itself.

Even when the source center is fixed, bubbling can separate its prescribed local data from the nonconstant sphere components; the analysis of bubbling goes back to Sacks and Uhlenbeck’s work on harmonic maps [40] and Gromov’s theory of pseudoholomorphic curves [21]. Since area may also concentrate near the source boundary, we must obtain a nonconstant sphere that satisfies both the required incidence conditions and the area bound.

1.2. Main results

For rational connectedness, we need a rational curve that meets a fiber of the rational quotient without remaining inside it. The main theorem therefore treats a compact fiber of a holomorphic map q defined on an open neighborhood N . Here $F = q^{-1}(y_0)$ denotes the fiber of q over y_0 in N , including all its components. We use the Ricci and area conventions in (1).

Theorem 1.1. *Let X be a complex Fano manifold of dimension n , equipped with a Kähler metric ω satisfying*

$$\mathrm{Ric}(\omega) \geq \kappa\omega, \quad \kappa > 0.$$

For an open subset $N \subset X$ and a complex manifold Y of positive dimension, consider a holomorphic map $q : N \rightarrow Y$ and a point $y_0 \in Y$. Suppose that the fiber $F := q^{-1}(y_0)$ is nonempty and compact and that q is a submersion on a neighborhood of F in N . Then there is a nonconstant holomorphic map $u : \mathbb{P}^1 \rightarrow X$ such that

$$u(\mathbb{P}^1) \cap F \neq \emptyset, \quad u(\mathbb{P}^1) \not\subset F, \quad \int_{\mathbb{P}^1} u^* \omega \leq \frac{2\pi(n+1)}{\kappa}.$$

Every Fano manifold admits such a Kähler metric by Yau's prescribed-Ricci theorem [50], as recalled in Proposition 3.1; the metric need not be Kähler–Einstein. Since X is projective, the reduced image of the sphere in the theorem is a rational curve.

This is an explicit metric area bound depending on the chosen Ricci lower bound κ .

The relative viewpoint has a classical antecedent in Campana's generic-fiber escape criterion and his Fano theorem [6, Proposition 2.7 and Theorem 3.1]. We work near a specified compact fiber of a holomorphic submersion and obtain a metric area bound.

During the construction, the center is allowed to move within F . If compactness has not already produced a sphere meeting F but not contained in F , this freedom allows the normalized base derivative to pass to the principal map. The final limit used for the explicit area bound retains the two incidence properties, though not necessarily the derivative normalization; hence the theorem prescribes neither a tangent direction nor a contact order.

Choose a holomorphic coordinate map q on a neighborhood of x . Its fiber over $q(x)$ is $F = \{x\}$, so the relative theorem gives the following pointwise statement.

Theorem 1.2. *Let X be a complex Fano manifold of dimension $n \geq 1$, equipped with a Kähler metric ω satisfying $\mathrm{Ric}(\omega) \geq \kappa\omega$ for some $\kappa > 0$. Then, for every $x \in X$, there is a nonconstant holomorphic map $u_x : \mathbb{P}^1 \rightarrow X$ such that*

$$x \in u_x(\mathbb{P}^1), \quad \int_{\mathbb{P}^1} u_x^* \omega \leq \frac{2\pi(n+1)}{\kappa}.$$

Classical bend-and-break produces a rational curve C through a prescribed point of a Fano n -fold with

$$0 < -K_X \cdot C \leq n + 1.$$

Mori's account [35, p. 749] attributes this form of the result to Kollár, by a technique similar to that of [34]. Integration over C gives

$$\kappa \int_C \omega \leq \int_C \mathrm{Ric}(\omega) = 2\pi(-K_X \cdot C).$$

The classical result controls the anticanonical degree, giving more information than the area bound for a fixed metric. The analytic construction controls area and weighted derivatives during the process. For the classical argument, see [8, Chapter 3].

1.3. The disk operations and their quantitative control

For a holomorphic map $f : \mathbb{D}_R \rightarrow X$, write

$$\text{Area}_R(f) = \int_{\mathbb{D}_R} f^* \omega, \quad B_R(f) = \sup_{z \in \mathbb{D}_R} (R - |z|) |f_z(z)|_\omega.$$

A bound $B_R(f) \leq B_0$ gives $|f_z(z)|_\omega \leq B_0/(R - |z|)$, and hence a derivative bound on every smaller disk. The quantity $B_R(f)$ is central to Brody's original proof of the reparametrization lemma [5, Lemma 2.1]: a point where the weighted derivative is nearly maximal supplies the center and scale for a map on a larger disk with controlled derivatives on compact subdisks. We use bounded values of B to choose uniform surgery parameters, and sequences with B tending to infinity to extract bubbles by point selection.

In the relative construction, we allow the disk center to move within F while fixing one nonzero scalar component of the derivative of $q \circ f$ there.

The first surgery lowers area while fixing the center and one scalar component of the derivative. Its geometric starting point is a paired second variation. For a disk extending holomorphically across its boundary, the variation fields are holomorphic sections ξ of f^*T_X , smooth on the closed disk. The Kähler condition identifies the sum of the second area derivatives along the real and imaginary parameter axes with the boundary quadratic form

$$\mathcal{H}_{R,f}(\xi) := \int_{\partial \mathbb{D}_R} \partial_\nu |\xi|_\omega^2 ds$$

whenever ξ is realized by a holomorphic family (Lemma 3.3), where ν is the outward Euclidean unit normal and ds is arclength. The paired variation therefore depends only on the field along the disk.

In the proof of Myers's theorem [36], Ricci curvature enters by tracing second variations built from a parallel orthonormal frame along a geodesic [33, §19, Theorem 19.4]; for holomorphic disks the analogous trace calculation uses a holomorphic frame $e = (e_1, \dots, e_n)$ of f^*T_X that is smooth on the closed disk and unitary on its boundary. Such a frame is supplied by Wiener–Masani spectral factorization [46], in the form used for Hermitian holomorphic bundles by Berndtsson and Rosay [3, Theorem 2.1 and Proposition 2.3]: their curvature-potential estimate, combined with the Green-potential bound below, controls the normalized frame and its inverse uniformly under fixed area and weighted-derivative bounds. These bounds feed the quantitative holomorphic realization of the variation fields; boundary unitarity is a normalization of the frame and imposes no geometric hypothesis on X .

Let G be the Gram matrix of this frame, with entries $G_{jk} = \langle e_j, e_k \rangle_\omega$. Since $G = \text{Id}_n$ on the boundary, $\partial_\nu \text{tr } G = \partial_\nu \log \det G$ there. Green's formula and the determinant-curvature identity therefore give, before imposing any constraint at the center,

$$\sum_{a=1}^n \mathcal{H}_{R,f}(e_a) = \int_{\partial \mathbb{D}_R} \partial_\nu \log \det G ds = -2 \int_{\mathbb{D}_R} f^* \text{Ric}(\omega).$$

Ricci curvature enters the deformation of a holomorphic disk through this identity; see (25) with $\Phi = \text{Id}_n$. Imposing the interpolation at the center adds a Dirichlet cost to it, and the scalar constraint means keeping one chosen component of $(q \circ f)_z(0)$ fixed. Write this component as $\lambda(f_z(0))$, where $\lambda \in T_{f(0)}^*X$ is the corresponding nonzero covector. A constant unitary change of frame then places $e_2(0), \dots, e_n(0)$ in $\ker \lambda$. Vanishing order at least two in the first component and at least one in the others then suffices to preserve the center and the specified scalar derivative in the affine families below. Taking monomials of the lowest admissible degrees $(2, 1, \dots, 1)$, normalized on $|z| = R$, gives Dirichlet cost $\pi(n+1)$ and paired trace

$$4\pi(n+1) - 2 \int_{\mathbb{D}_R} f^* \text{Ric}(\omega) \leq 4\pi(n+1) - 2\kappa \text{Area}_R(f).$$

It follows that, above $A_* = 2\pi(n+1)/\kappa$, the boundary quadratic form is negative on at least one of these fields. The realization below turns this negativity into area decrease. The calculation after (29) proves that this Dirichlet cost is the least possible under the prescribed interpolation and boundary conditions.

Other choices of field are available: using $(z/R)^m$ in every direction, for a fixed integer $m \geq 2$, gives paired trace $4\pi mn - 2 \int_{\mathbb{D}_R} f^* \text{Ric}(\omega)$ and threshold $2\pi mn/\kappa$. The perturbation then vanishes to order m and preserves the entire $(m-1)$ -jet at the center, while $m=2$ preserves the first jet, at threshold $4\pi n/\kappa$. These statements concern the disk deformations only, and not the jets of the final sphere.

To obtain uniform deformations from this calculation, we lift the disk and its fields to a fixed affine Stein manifold lying over X . Quantitative corona estimates [37] control the lifts; Euclidean perturbation followed by a holomorphic neighborhood retraction [11, 16] and projection to X gives actual holomorphic families. Radial regularization makes the fields extend across the source boundary and preserves the center constraints exactly. For the resulting families, the sum of the second area derivatives along the real and imaginary parameter axes differs from the trace calculated above by an error that can be made arbitrarily small. Fix area and weighted-derivative bounds A_0, B_0 and require $\text{Area}_R(f) \geq A_* + \delta$ for fixed $\delta > 0$. This sum remains negative for the realized families, so at least one real parameter direction has negative second variation. We choose its sign to make the first-order contribution nonpositive. A common parameter interval and cubic remainder now give a positive step t_* and an area decrease $\sigma > 0$, uniform in the disk and its radius. These constants depend on the fixed bounds and the margin δ ; see Proposition 2.1. This uniform decrease provides room for the area cost of enlargement.

For the second surgery, choose a holomorphic lift H of f into an affine space and a holomorphic map Π back to X , defined near the lifted image, with $\Pi \circ H = f$. For sufficiently small $t > 0$, set

$$h_t(z) = \Pi(H(z) + tzH'(0)), \quad G_t(z) = h_t(z/(1+t)).$$

The affine perturbation fixes the center and multiplies $f_z(0)$ by $1+t$. Rescaling the source then restores the original derivative and increases the source radius from R to $(1+t)R$. On these respective source disks, both the area and the squared weighted derivative change by at most a uniform constant times t under fixed area and weighted-derivative bounds (Proposition 2.2).

Alternating enlargement and area reduction keeps the area below a fixed cap, with quantitative control of each finite stage in Proposition 2.9. If derivatives concentrate while the source radii remain bounded, compactness either produces the required sphere or retains a normalized principal map. In the latter case, a bubble accounts for a definite area loss, and we enlarge the principal map before continuing. The recovery argument in Section 2.4 organizes these returns and yields the required sphere or disks of arbitrarily large radius; this passage still uses subsequential limits.

We obtain the stated bound by letting the area cap decrease to A_* . During this passage, we mark one point mapping into F and another mapping outside a fixed neighborhood of F . The limit may have several sphere components; the two markings allow us to select one that meets F without being contained in it, with area at most A_* . The compactness argument is given in Section 5, using the theorem of Ivashkovich and Shevchishin [28].

The controlled holomorphic deformations developed here also suggest an approach to Conjecture A, in which deformations of disks would lead to dominating holomorphic sprays on Fano manifolds.

Plan of the paper.

Section 2 proves the relative sphere theorem and records the value of the bound for the Fubini–Study metric. The Ricci-trace calculation, uniform holomorphic realization, and marked compactness are established in Sections 3–5. The final section recalls how the construction combines with classical algebraic arguments to recover rational connectedness and Mori’s ample-tangent characterization of projective space.

2. EXPLICIT CONSTRUCTION OF A RELATIVE SPHERE

2.1. Normalized disks and the two operations

We fix a complex Fano manifold X of dimension $n \geq 1$ and a Kähler metric ω satisfying $\text{Ric}(\omega) \geq \kappa \omega$ for some $\kappa > 0$ throughout the construction. In holomorphic coordinates $w^1, \dots, w^n$, set $h_{\alpha\bar{\beta}} = \langle \partial/\partial w^\alpha, \partial/\partial w^\beta \rangle_\omega$ for the pairwise inner products of the coordinate vector fields; their Gram matrix is $h = (h_{\alpha\bar{\beta}})$. Our Hermitian inner products are complex-linear in the first variable and conjugate-linear in the second [9, Ch. V, §12]. Coefficient columns a, b pair as $a^\top h \bar{b}$, where $\top$ denotes ordinary transpose. For a holomorphic source map f , write $f_z = df(\partial/\partial z)$. With repeated coordinate indices summed and dA denoting Euclidean area measure, our normalizations are

$$\omega = \frac{i}{2} h_{\alpha\bar{\beta}} dw^\alpha \wedge d\bar{w}^\beta, \quad f^* \omega = |f_z|_h^2 dA, \quad \text{Ric}(\omega) = -i\partial\bar{\partial} \log \det h. \quad (1)$$

For a holomorphic map $f : S \rightarrow X$ from a Riemann surface, set $\text{Area}(f) := \int_S f^* \omega$; this counts geometric area with the multiplicity of the map. A holomorphic sphere is a map $u : \mathbb{P}^1 \rightarrow X$. When u is nonconstant, we call its reduced image a rational curve, while $\text{Area}(u)$ continues to include the mapping multiplicity. For $a \in \mathbb{C}$ and $r > 0$, write $\mathbb{D}_r(a) := \{z \in \mathbb{C} : |z - a| < r\}$, and abbreviate $\mathbb{D}_R := \mathbb{D}_R(0)$ and $\mathbb{D} := \mathbb{D}_1(0)$. We write $\bar{\mathbb{D}}_R = \{z \in \mathbb{C} : |z| \leq R\}$ for the closed disk and $\partial\mathbb{D}_R = \{z \in \mathbb{C} : |z| = R\}$ for its boundary circle. We use the area on $\mathbb{D}_R$ and the boundary-distance-weighted derivative bound

$$\text{Area}_R(f) := \int_{\mathbb{D}_R} f^* \omega, \quad B_R(f) := \sup_{z \in \mathbb{D}_R} (R - |z|) |f_z(z)|_\omega. \quad (2)$$

Write $\mathcal{O}(V, X)$ for the holomorphic maps from an open set $V \subset \mathbb{C}$ to X , and $K \Subset U$ when the closure of K is compact and contained in U . Smooth convergence means uniform convergence of all derivatives on compact source subsets.

For this fixed Ricci lower bound, set

$$A_* = \frac{2\pi(n+1)}{\kappa}. \quad (3)$$

The Fano manifold X is projective. We fix an embedding $\iota : X \hookrightarrow \mathbb{P}^{m-1}$ to carry out the two disk operations. It remains fixed as the map and source radius vary, and the constants may depend on it. All areas and target derivative norms are still measured using ω .

For the relative constraint, start with $q : N \rightarrow Y$, y_0 , and the compact fiber $F = q^{-1}(y_0)$ in Theorem 1.1. Choose a smooth-boundary neighborhood U with $F \Subset U \Subset N$, small enough that q is a submersion on a neighborhood of $\bar{U}$. Since $F = q^{-1}(y_0) \Subset U$, one has $y_0 \notin q(\partial U)$, and $q(U)$ is an open neighborhood of y_0 . We can therefore choose a coordinate ball Y' about y_0 such that $\bar{Y}' \cap q(\partial U) = \emptyset$ and $Y' \subset q(U)$. After replacing N by $U \cap q^{-1}(Y')$ and Y by Y' , the map q is a proper holomorphic submersion: it remains a submersion by the choice of U , and the inverse image of a compact subset of Y' is closed in $\bar{U}$ and avoids ∂U . In the restricted coordinate ball write $d_Y = \dim Y$ and let $\psi : N \rightarrow \mathbb{C}^{d_Y}$ be the coordinate expression of q , translated so that $\psi^{-1}(0) = F$. Fix $0 \neq \eta \in (\mathbb{C}^{d_Y})^*$ and set

$$\lambda^F = (\lambda_x^F)_{x \in F}, \quad \lambda_x^F := \eta \circ d\psi_x \in T_x^* X.$$

The covectors λ_x^F are the differentials of the same scalar function $\eta \circ \psi$ at points $x \in F$, so even when $f(0)$ moves in F , normalization always means $(\eta \circ \psi \circ f)_z(0) = 1$, with the composition defined near zero. It follows that $d\psi_{f(0)}(f_z(0)) \neq 0$, so the initial tangent vector is not tangent to F . The submersion property makes λ_x^F nonzero for every $x \in F$. The classes below consist of maps on $\mathbb{D}_R$ that extend holomorphically to some open neighborhood of $\bar{\mathbb{D}}_R$. The neighborhood may depend on the map. Set

$$\mathcal{A}_R^{F, \lambda^F} = \left\{ f \in \mathcal{O}(\mathbb{D}_R, X) : f(0) \in F, \quad \lambda_{f(0)}^F(f_z(0)) = 1 \right\}. \quad (4)$$

We call a holomorphic disk *normalized* if its center lies in F and its base scalar derivative equals 1. Membership in $\mathcal{A}_R^{F, \lambda^F}$ requires, in addition, holomorphic extension across the source boundary. In the iteration, A and B always refer to the current map and its current source radius.

Choose $x_0 \in F$ and a holomorphic coordinate chart χ with $\chi(x_0) = 0$. Since $\lambda_{x_0}^F$ is nonzero, choose $v \in \mathbb{C}^n$ such that $\lambda_{x_0}^F(d(\chi^{-1})_0(v)) = 1$. For all sufficiently small $R_0 > 0$, the map $f_0(z) = \chi^{-1}(zv)$ is holomorphic near $\mathbb{D}_{R_0}$ and belongs to $\mathcal{A}_{R_0}^{F, \lambda^F}$. Moreover, $\text{Area}_{R_0}(f_0) \rightarrow 0$ as $R_0 \downarrow 0$. Fix one such R_0 with $\text{Area}_{R_0}(f_0) < A_*$; the pair (f_0, R_0) is the initial disk for the iteration.

To relate the source radius to the weighted derivative, let

$$\ell_F := \max_{x \in F} \|\lambda_x^F\|_{\omega, *},$$

the maximal dual norm of the covector field λ^F along F . Here $\|\cdot\|_{\omega, *}$ is the dual norm induced by ω . Since F is compact and $\lambda_x^F \neq 0$ on F , we have $0 < \ell_F < \infty$. For every normalized disk, the scalar condition gives $1 = |\lambda_{f(0)}^F(f_z(0))| \leq \ell_F |f_z(0)|_\omega$. Evaluating the supremum defining $B_R(f)$ at $z = 0$, we obtain

$$B_R(f) \geq R |f_z(0)|_\omega \geq R / \ell_F. \quad (5)$$

Each surgery keeps the value at 0 and the normalized scalar derivative fixed. When a limiting disk is retained for the next stage, its value at 0 is still in F , but may be a different point of F ; its scalar derivative is still 1. The example in Section 5.3 illustrates this with $F = \mathbb{P}^1 \times \{[0 : 1]\}$. The new maps need not agree with their predecessors on the original source disks: enlargement is achieved by deformation, not by analytic continuation of a single map. Fix also $F \Subset V \Subset N$ with smooth boundary.

Fix $c > 0$ with $\omega \geq c \iota^* \omega_{\text{FS}}$, normalizing a projective line to have Fubini–Study area π , and set

$$\hbar_X = c\pi > 0. \quad (6)$$

If $u : \mathbb{P}^1 \rightarrow X$ is nonconstant, the projective map $\iota \circ u$ has an integer degree $d \geq 1$. Hence

$$\text{Area}(u) \geq c \int_{\mathbb{P}^1} u^* \iota^* \omega_{\text{FS}} = c\pi d \geq \hbar_X.$$

The two operations play complementary roles. Surgery I acts on a fixed source disk and lowers its area above the Ricci threshold, using the Kähler variation identity of Lemma 3.3. Surgery II enlarges the source at a linear area cost. The next proposition makes the first operation uniform and records its effect on B_R .

Proposition 2.1. *In the relative setting fixed above, let $A_0, B_0, \delta > 0$. There are constants $\sigma > 0$ and $\beta_I < \infty$, depending only on these three bounds and the fixed relative data, with the following property. For every $R > 0$, each $f \in \mathcal{A}_R^{F, \lambda^F}$ with*

$$A_* + \delta \leq \text{Area}_R(f) \leq A_0, \quad B_R(f) \leq B_0$$

admits a map $g \in \mathcal{A}_R^{F, \lambda^F}$ satisfying

$$g(0) = f(0), \quad \text{Area}_R(g) \leq \text{Area}_R(f) - \sigma, \quad |B_R(g)^2 - B_R(f)^2| \leq \beta_I. \quad (7)$$

The last estimate gives $B_R(g) \leq (B_0^2 + \beta_I)^{1/2}$, so Surgery I controls B_R without necessarily decreasing it. The map g is explicit: it belongs to the affine-lift family of Theorem 4.8, and (60) records its step size and guaranteed area decrease.

The second surgery perturbs a lifted disk in affine space, projects back to X , and dilates the source.

Proposition 2.2. *Let (X, ω) be a compact projective manifold with a smooth Hermitian metric, and fix an embedding $\iota : X \hookrightarrow \mathbb{P}^{m-1}$. Fix bounds $A_0, B_0 \geq 0$. There are constants $0 < \bar{t} \leq 1$ and $C_{II} < \infty$, independent of the radius and the individual map, with the following properties.*

- (i) *Let $R > 0$ and let h be a map to X , holomorphic on a neighborhood of $\mathbb{D}_R$, with $\text{Area}_R(h) \leq A_0$ and $B_R(h) \leq B_0$. There are a holomorphic lift $H : U \rightarrow \mathbb{C}^{2m}$ on a neighborhood U of $\mathbb{D}_R$ and a holomorphic ambient projection Π defined near $H(\mathbb{D}_R)$, with $\Pi \circ H = h$ near $\mathbb{D}_R$, such that the formulas*

$$h_t(z) = \Pi(H(z) + tzH'(0)), \quad G_t(z) = h_t(z/(1+t)), \quad 0 < t \leq \bar{t}, \quad (8)$$

give holomorphic maps $h_t : \mathbb{D}_R \rightarrow X$ and $G_t : \mathbb{D}_{(1+t)R} \rightarrow X$ that extend past their respective closed disks. The map G_t satisfies

$$|\text{Area}_{(1+t)R}(G_t) - \text{Area}_R(h)| \leq C_{II}t, \quad (9)$$

$$|B_{(1+t)R}(G_t)^2 - B_R(h)^2| \leq C_{II}t/\pi. \quad (10)$$

The same estimates compare h_t and h on $\mathbb{D}_R$, and $(h_t)_z(0) = (1+t)h_z(0)$, while $G_t(0) = h(0)$ and $(G_t)_z(0) = h_z(0)$.

- (ii) *Let h be holomorphic only on the finite open disk $\mathbb{D}_R$, with the same area and derivative bounds. For each $0 < t \leq \bar{t}$, let $r = (1+t/2)R/(1+t)$ and apply part (i) to $h|_{\mathbb{D}_r}$. If H_r is the resulting lift, then*

$$R' = (1+t)r = (1+t/2)R, \quad g(z) = \Pi \left(H_r \left(\frac{z}{1+t} \right) + t \frac{z}{1+t} H'_r(0) \right) \quad (z \in \mathbb{D}_{R'}) \quad (11)$$

extends holomorphically past $\mathbb{D}_{R'}$. Since $r < R < R'$, its source is larger than that of h , and

$$\text{Area}_{R'}(g) \leq \text{Area}_R(h) + C_{II}t, \quad B_{R'}(g)^2 \leq B_R(h)^2 + C_{II}t/\pi. \quad (12)$$

Moreover, $g(0) = h(0)$ and $g_z(0) = h_z(0)$.

Propositions 2.1 and 2.2 are proved in Section 4.4.

In (8), the affine perturbation changes the derivative on the fixed source disk, whereas the subsequent dilation enlarges the source and restores the original derivative without changing the perturbed image or its area.

2.2. Principal limits and marked values

We first construct a principal map in the original source coordinate, then a rational chain retaining two marked values. If the principal map is entire and has finite area, removal at infinity extends it to a sphere. We derive these statements from the compactness and removal results of Ivashkovich and Shevchishin [28]. Proposition 2.3, Theorem 2.4, and Lemma 2.5 are proved in Section 5.

Proposition 2.3. *Let (X, ω) be a compact Hermitian manifold. For each j , let $R_j > 0$ and let $f_j : U_j \rightarrow X$ be holomorphic, where $U_j \subset \mathbb{C}$ is open and $\mathbb{D}_{R_j} \Subset U_j$. Assume that $\sup_j \text{Area}_{R_j}(f_j) < \infty$, and set $A_\infty = \liminf_j \text{Area}_{R_j}(f_j)$. Choose $R_\infty \in (0, \infty]$ with $R_\infty \leq \liminf_j R_j$, interpreting $\mathbb{D}_\infty$ as $\mathbb{C}$. After passing to a subsequence realizing A_∞ and then to a further subsequence, there are a finite set $\Sigma_\infty \subset \mathbb{D}_{R_\infty}$ and a holomorphic map $f_\infty : \mathbb{D}_{R_\infty} \rightarrow X$ such that, in the original source coordinate,*

$$f_j \longrightarrow f_\infty \quad \text{smoothly on compact subsets of } \mathbb{D}_{R_\infty} \setminus \Sigma_\infty, \quad \text{Area}_{R_\infty}(f_\infty) \leq A_\infty.$$

The possibly constant map f_∞ is holomorphic across Σ_∞ .

The extension across Σ_∞ does not assert convergence of f_j at the concentration points.

Theorem 2.4. *Let (X, ω) be a compact Hermitian manifold. Every holomorphic map from a punctured disk to X with finite ω -area extends across the puncture. Every finite-area holomorphic map $u : \mathbb{C} \rightarrow X$ extends to a holomorphic map $\mathbb{P}^1 \rightarrow X$.*

Lemma 2.5. *Let (X, ω) be a compact Hermitian manifold, and let $K_0, K_1 \subset X$ be disjoint compact subsets. Suppose $R_j > 1$, $R_j \rightarrow \infty$, and the holomorphic maps $f_j : U_j \rightarrow X$, where $U_j \subset \mathbb{C}$ is open and $\overline{\mathbb{D}}_{R_j} \in U_j$, satisfy*

$$f_j(0) \in K_0, \quad f_j(1) \in K_1, \quad \sup_j \text{Area}_{R_j}(f_j) < \infty.$$

Then there are nonconstant holomorphic maps $u_1, \dots, u_\ell : \mathbb{P}^1 \rightarrow X$ whose images form a chain: the first meets K_0 , the last meets K_1 , and consecutive images intersect. Their total area, counted with the multiplicities of these maps, satisfies

$$\sum_{i=1}^{\ell} \text{Area}(u_i) \leq \liminf_j \text{Area}_{R_j}(f_j).$$

The disjoint target sets force the path between the two marked values to contain a nonconstant component. Equality of nodal values preserves incidence when constant components along that path are omitted.

The scalar normalization is imposed on the local base function: $(\eta \circ \psi \circ f)_z(0) = 1$. If first-exit points from a fixed neighborhood of F approach zero, rescaling and marked compactness give a sphere meeting F but not contained in F . Otherwise, a subsequence remains in that neighborhood on a fixed source disk, where the functions $\eta \circ \psi \circ f_j$ are uniformly bounded. Cauchy estimates then pass their derivatives at zero to the principal limit.

Lemma 2.6. *Let (X, ω) be a compact Hermitian manifold, and assume the relative setting fixed above: $F \subset N \subset X$ is the nonempty compact fiber of the proper holomorphic submersion $q : N \rightarrow Y$, ψ is its local coordinate expression with $F = \psi^{-1}(0)$, and $\lambda_x^F = \eta \circ d\psi_x$ for the fixed $\eta \neq 0$. Suppose $R_j \rightarrow R_\infty \in (0, \infty]$, where $\mathbb{D}_\infty = \mathbb{C}$, and $f_j \in \mathcal{A}_{R_j}^{F, \lambda^F}$ (see (4)), with*

$$\text{Area}_{R_j}(f_j) \rightarrow E < \infty.$$

Then at least one of the following conclusions holds:

(i) *There is a nonconstant sphere $v : \mathbb{P}^1 \rightarrow X$ such that*

$$v(\mathbb{P}^1) \cap F \neq \emptyset, \quad v(\mathbb{P}^1) \not\subset F, \quad \text{Area}(v) \leq E.$$

(ii) *A subsequence has a principal limit $h : \mathbb{D}_{R_\infty} \rightarrow X$ in the original source coordinate, as in Proposition 2.3, satisfying*

$$\text{Area}_{R_\infty}(h) \leq E, \quad h(0) \in F, \quad \lambda_{h(0)}^F(h_z(0)) = 1.$$

On some fixed disk about zero, the base compositions $\psi \circ f_j$ converge to $\psi \circ h$ smoothly on compact subsets.

Proof. Choose $F \Subset V \Subset N$ with smooth boundary, and fix $r > 0$ such that $R_j > r$ for all sufficiently large j . There are two possibilities after passage to a subsequence.

First, suppose there are $z_j \rightarrow 0$ with $z_j \neq 0$ and $f_j(z_j) \in \partial V$. Such points can be chosen as first exits along segments from zero whenever no fixed small closed source disk has image in V along an infinite subsequence. The rescaled maps

$$v_j(\zeta) := f_j(z_j \zeta), \quad |\zeta| < R_j/|z_j|,$$

extend past their closed source disks and satisfy

$$v_j(0) \in F, \quad v_j(1) \in \partial V, \quad \text{Area}_{R_j/|z_j|}(v_j) = \text{Area}_{R_j}(f_j) \rightarrow E.$$

Their source radii tend to infinity. Lemma 2.5 gives a chain of nonconstant spheres from F to ∂V , of total area at most E . Take its first component not contained in F . It meets F , either at the first marked value or at its intersection with the preceding component, and has area at most E . This proves (i).

Otherwise, for some $0 < \rho < r$, an infinite subsequence satisfies $f_j(\overline{\mathbb{D}}_\rho) \subset V$. Apply Proposition 2.3 on $\mathbb{D}_{R_\infty}$ to this subsequence. Its principal map h takes values in $\overline{V}$ on $\mathbb{D}_\rho$: this holds away from the finite concentration set by convergence, and at that set by continuity. In particular, $\psi \circ h$ is defined there. Since $\overline{V} \Subset N$, the holomorphic maps $\psi \circ f_j$ are uniformly bounded on $\mathbb{D}_\rho$. A further subsequence converges locally uniformly to a holomorphic map, which equals $\psi \circ h$ away from the finite concentration set by principal convergence and hence throughout $\mathbb{D}_\rho$ by the identity theorem. Cauchy estimates give convergence of all derivatives of these base maps on compact subdisks, even if 0 is a concentration point. This does not exclude a bubble at 0 inside a fiber of ψ . Since $f_j(0) \in F$ and $(\eta \circ \psi \circ f_j)_z(0) = 1$, we obtain

$$\psi(h(0)) = 0, \quad (\eta \circ \psi \circ h)_z(0) = \lim_j (\eta \circ \psi \circ f_j)_z(0) = 1.$$

The area bound follows from principal compactness, proving (ii). $\square$

The following standard consequence of local small-energy regularity supplies the weighted derivative bound for a principal map on a finite disk; we include the short argument, since the resulting bound may depend on the individual map.

Lemma 2.7. *Let (X, ω) be a compact Hermitian manifold, let $0 < R < \infty$, and let $h : \mathbb{D}_R \rightarrow X$ be holomorphic with $\text{Area}_R(h) < \infty$. Then $B_R(h) < \infty$.*

The proof in Section 5.4 applies the small-energy estimate of Ivashkovich and Shevchishin [28, Lemma 1.1] in a boundary collar.

Consequently, whenever compactness retains a principal map h on a finite disk, one may set $B_0 = B_R(h)$ and apply Proposition 2.2 with this derivative bound and the current area bound. The enlargement parameters are chosen only after h is known and may therefore vary from one principal map to the next. The two-sided B^2 estimate (10) applies to a closed-disk enlargement; when h is defined only on its open disk, we use the repair and the upper bound in (12).

2.3. Recovery with an area loss

Lemma 2.8. *In the relative setting fixed above, let $R_j > 0$ and let $f_j : \mathbb{D}_{R_j} \rightarrow X$ extend holomorphically past $\overline{\mathbb{D}}_{R_j}$, with $f_j(0) \in F$ and $\lambda_{f_j(0)}^F((f_j)_z(0)) = 1$. Assume $R_j \rightarrow R \in (0, \infty)$, $\text{Area}_{R_j}(f_j) \rightarrow E < \infty$, and $B_{R_j}(f_j) \rightarrow \infty$. One may select either a nonconstant sphere $v : \mathbb{P}^1 \rightarrow X$ satisfying*

$$v(\mathbb{P}^1) \cap F \neq \emptyset, \quad v(\mathbb{P}^1) \not\subset F, \quad \text{Area}(v) \leq E,$$

or a normalized principal map $h : \mathbb{D}_R \rightarrow X$ satisfying

$$\text{Area}_R(h) \leq E - \hbar_X, \quad B_R(h) < \infty. \quad (13)$$

The same conclusion holds if the f_j are holomorphic only on the open disks $\mathbb{D}_{R_j}$, without extension across their boundaries; all other hypotheses are unchanged.

Proof. Apply Lemma 2.6. If its first alternative gives the required sphere, we are done. Otherwise, pass to the subsequence in its second alternative. It has a normalized principal map $h : \mathbb{D}_R \rightarrow X$ with $\text{Area}_R(h) \leq E$ and a finite concentration set Σ . We use the same subsequence below, so its areas still tend to E and its weighted derivatives still diverge.

Apply Lemma 5.1 to this subsequence. In its construction, $w_j \in \mathbb{D}_{R_j}$ maximizes the weighted derivative. Put $d_j = R_j - |w_j|$ and $\Lambda_j = |(f_j)_z(w_j)|_\omega$. Then

$$d_j \Lambda_j = B_{R_j}(f_j) \longrightarrow \infty, \quad \Lambda_j \geq B_{R_j}(f_j)/R_j \longrightarrow \infty.$$

The rescaled maps $u_j(\zeta) = f_j(w_j + \zeta/\Lambda_j)$ converge locally smoothly on $\mathbb{C}$ to a nonconstant holomorphic map u . Pass to a further subsequence with $w_j \rightarrow w \in \overline{\mathbb{D}}_R$. If w were in $\mathbb{D}_R \setminus \Sigma$, the smooth convergence to h near w would bound Λ_j . Thus $w \in \Sigma \cup \partial \mathbb{D}_R$.

We now count the areas of h and u on disjoint parts of the same source disks. Fix $S > 0$ and a compact region $Q \subset \mathbb{D}_R \setminus \Sigma$. For large j , the disk $\Delta_{j,S} = \mathbb{D}_{S/\Lambda_j}(w_j)$ lies in $\mathbb{D}_{R_j}$ because $d_j \Lambda_j \rightarrow \infty$. It shrinks to w , so it is disjoint from Q . The compact set Q is also contained in $\mathbb{D}_{R_j}$ for large j . Hence

$$\int_Q f_j^* \omega + \int_{\Delta_{j,S}} f_j^* \omega \leq \text{Area}_{R_j}(f_j).$$

The first integral converges to $\int_Q h^* \omega$ by smooth principal convergence. The source change $z = w_j + \zeta/\Lambda_j$ gives $\int_{\Delta_{j,S}} f_j^* \omega = \text{Area}_S(u_j) \rightarrow \text{Area}_S(u)$. Thus

$$\int_Q h^* \omega + \text{Area}_S(u) \leq E.$$

Exhaust $\mathbb{D}_R \setminus \Sigma$ by such regions Q , then let $S \rightarrow \infty$. The finite set Σ contributes no area to the holomorphic map h , so

$$\text{Area}_R(h) + \text{Area}(u) \leq E.$$

This also covers $w \in \partial \mathbb{D}_R$: every fixed Q remains a positive distance from w . The map u has finite area and thus extends to a nonconstant sphere by Theorem 2.4. The projective degree estimate following (6) gives $\text{Area}(u) \geq \hbar_X$. Therefore $\text{Area}_R(h) \leq E - \hbar_X$. Finally, Lemma 2.7 gives $B_R(h) < \infty$.

This proves the closed-disk assertion. For the open-disk assertion in the last sentence of the lemma, the maps need not extend across $\partial \mathbb{D}_{R_j}$. Their areas converge to $E < \infty$, so Lemma 2.7 gives $B_{R_j}(f_j) < \infty$ for all large j . Choose $w_j \in \mathbb{D}_{R_j}$ with $(R_j - |w_j|)|(f_j)_z(w_j)|_\omega \geq B_{R_j}(f_j)/2$, and put $d_j = R_j - |w_j|$. Since the area of each disk is finite, we can choose $r_j < R_j$ with

$$R_j - r_j < \min\{1/j, d_j/2\}, \quad \text{Area}_{R_j}(f_j) - \text{Area}_{r_j}(f_j) < 1/j.$$

Then $r_j > |w_j|$, and the restriction to $\mathbb{D}_{r_j}$ extends past its closed disk. Moreover,

$$B_{r_j}(f_j) \geq (r_j - |w_j|)|(f_j)_z(w_j)|_\omega \geq \frac{1}{4}B_{R_j}(f_j) \rightarrow \infty.$$

The restrictions retain the center and scalar normalization; their radii tend to R and their areas tend to E . Thus the closed-disk case just proved applies. $\square$

In the principal-map alternative, recovery keeps the whole principal map, discards the rescaled sphere, and allows the center to move in F ; it is a subsequential operation on a sequence of disks, not a cutting-and-gluing operation on one given disk, and it therefore carries no closeness assertion near a bubble.

2.4. Iteration and the relative sphere

Fix $\delta > 0$ and set

$$L = A_* + \delta, \quad \eta_0 = \delta/2, \quad C = A_* + 2\delta. \quad (14)$$

For a stage threshold $M > 0$, choose all constants with the same area and derivative caps C, M . Use the descent step σ from (60) and set

$$\tau_M = \min \left\{ \frac{\bar{t}}{2}, \frac{\eta_0}{\max\{1, C_{II}\}} \right\} > 0. \quad (15)$$

One enlargement multiplies the radius by $1 + \tau_M$ and adds at most η_0 to the area. Since $L + \eta_0 = A_* + 3\delta/2 < C$, enlarging a disk of area below L keeps it below the cap C . Apply the first applicable rule, restarting with the updated disk after each surgery:

- (1) If $B_R(f) \geq M$, record the disk and end the stage.
- (2) If $B_R(f) < M$ and $\text{Area}_R(f) \geq L$, perform Surgery I.
- (3) If $B_R(f) < M$ and $\text{Area}_R(f) < L$, perform Surgery II with $t = \tau_M$.

Proposition 2.9. *In the relative setting fixed above, let $\delta > 0$ and $M > 0$. Define L, η_0, C by (14), choose the surgery constants for the caps (C, M) , and set σ, τ_M as in (60) and (15). Let the starting disk $f_s \in \mathcal{A}_{R_s}^{F, \lambda^F}$ have radius $R_s > 0$ and area $A_s = \text{Area}_{R_s}(f_s) < C$. The preceding rules produce a disk with $A < C$ and $B \geq M$ in finitely many surgeries. If $B_{R_s}(f_s) < M$, then (5) gives*

$$R_s/\ell_F \leq B_{R_s}(f_s) < M.$$

Hence $R_s < \ell_F M$, so the numerator below is positive and $N_M \geq 1$. Set

$$N_M = \left\lceil \frac{\log(\ell_F M / R_s)}{\log(1 + \tau_M)} \right\rceil, \quad (16)$$

$$J_0 = \begin{cases} 0, & A_s < L, \\ \lfloor (A_s - L)/\sigma \rfloor + 1, & A_s \geq L, \end{cases} \quad (17)$$

$$J = \lfloor \eta_0/\sigma \rfloor + 1. \quad (18)$$

An explicit upper bound for the total number of surgeries is

$$T_M = J_0 + N_M(1 + J). \quad (19)$$

If the starting disk already has $B \geq M$, no surgery is required.

Proof. Both surgeries preserve normalization. Descent lowers area, and enlargement starts below L and ends below $L + \eta_0 < C$. Thus $A < C$ throughout. Each operation starts with $B < M$, so the chosen constants apply until the stage ends.

At most J_0 initial descents bring $A < L$ or end the stage. After each enlargement, at most J descents do the same; the extra 1 in the counts covers equality at L . Allow at most N_M such blocks. At the N_M th enlargement, if needed,

$$R_s(1 + \tau_M)^{N_M} \geq \ell_F M,$$

so (5) gives $B \geq M$. This proves the bound T_M . Summing the area changes over any completed prefix, with N_I, N_{II} denoting its surgery counts, also gives

$$0 \leq A \leq A_s - N_I \sigma + N_{II} \eta_0, \quad N_I \sigma \leq A_s + N_{II} \eta_0. \quad (20)$$

□

A finite hierarchy of principal recoveries.

A finite stage can make B arbitrarily large without forcing the source radius to grow beyond a prescribed value. At a bounded limiting radius, Lemma 2.8 either gives the required sphere or returns a principal map whose area has dropped by at least $\hbar_X$. The principal map may be defined only on an open disk. Repair makes it eligible for another finite stage but can increase its area; we therefore record each principal map as the area record *before* repairing it. The argument bounds only how deeply the recoveries nest; it does not bound the total number of finite stages or subsequence selections.

Proposition 2.10. *In the relative setting fixed above, let $\delta > 0$ and $C = A_* + 2\delta$ as in (14), and let $\hbar_X$ be the area quantum in (6). For every finite $T > 0$ and every $f \in \mathcal{A}_R^{F, \lambda^F}$ with $R > 0$ and $\text{Area}_R(f) < C$, the finite stages and principal recoveries give either a nonconstant sphere $v : \mathbb{P}^1 \rightarrow X$ satisfying $v(\mathbb{P}^1) \cap F \neq \emptyset$ and $v(\mathbb{P}^1) \not\subset F$, or a normalized disk of radius at least T . The resulting sphere or disk has area at most C , and source radii never decrease. At most $\lfloor C/\hbar_X \rfloor + 1$ recovery levels are nested. The resulting disk may be defined only on its open source disk.*

Proof. We first describe one recovery level. If the starting radius is at least T , there is nothing to do. Otherwise apply Proposition 2.9 successively with derivative thresholds $M = 1, 2, \dots$, stopping if a radius reaches T . If this never happens, the resulting closed-disk maps have areas below C ,

nondecreasing radii tending to some $R_* \leq T$, and weighted derivatives tending to infinity. Pass to a subsequence with convergent areas. Lemma 2.8 gives either the required sphere or a normalized principal map $h : \mathbb{D}_{R_*} \rightarrow X$ with $\text{Area}_{R_*}(h) \leq C - \hbar_X$. If $R_* = T$, this disk is a required output. One level therefore either completes the proposition or returns a principal disk of radius less than T and area at most $C - \hbar_X$.

We now iterate this alternative. More precisely, let P_k be the assertion that, starting from any normalized closed-disk map of area less than C , at most k nested recovery levels give either a required sphere or disk, or a normalized principal disk $h : \mathbb{D}_{R_h} \rightarrow X$ with

$$R_h < T, \quad \text{Area}_{R_h}(h) \leq C - k\hbar_X, \quad B_{R_h}(h) < \infty.$$

Source radii do not decrease. The preceding paragraph proves P_1 ; the finiteness of $B_{R_h}(h)$ follows from Lemma 2.7.

Assume P_k . Apply it to the starting disk. There is nothing further to prove if it gives a required sphere or disk. Otherwise denote its principal output by (h_1, R_1) . We may repeat the following operation as long as the output radius remains below T . The principal outputs have finite B , but no common bound for it is known. We therefore prepare the repair constants for every possible integer derivative cap: for each $q \geq 1$, take $\bar{t}_q$ and $C_{II,q}$ from Proposition 2.2 with area cap C and B -cap q , and set

$$t_q = \min \left\{ \bar{t}_q/2, \frac{\hbar_X}{2 \max\{1, C_{II,q}\}} \right\} > 0.$$

Given a principal output (h_j, R_j) , put $b_j = B_{R_j}(h_j)$ and $q_j = \max\{1, \lceil b_j \rceil\}$. Then $B_{R_j}(h_j) \leq q_j$ and $\text{Area}_{R_j}(h_j) \leq C$, so the repair estimates chosen for the caps (C, q_j) apply to h_j . Apply the open-disk repair (11) to h_j with $t = t_{q_j}$. It produces a normalized map g_j holomorphic past its closed source disk, with radius $(1 + t_{q_j}/2)R_j$ and

$$\text{Area}(g_j) \leq C - k\hbar_X + \hbar_X/2 < C.$$

If this radius reaches T , retain g_j . Otherwise apply P_k to g_j . A required sphere or disk ends the construction; in the remaining case record the next principal output (h_{j+1}, R_{j+1}) , again *before* repairing it. Its area is at most $C - k\hbar_X$, and

$$R_{j+1} \geq (1 + t_{q_j}/2)R_j.$$

Suppose this repetition continues indefinitely without reaching T . Then R_j increases to some $R_* \leq T$. We claim that $b_j \rightarrow \infty$. For each integer $Q \geq 1$, set $a_Q = \min_{1 \leq q \leq Q} t_q/2 > 0$. Every index with $b_j \leq Q$ contributes a radius factor of at least $1 + a_Q$. Since all radii stay below T ,

$$\#\{j : b_j \leq Q\} \leq \frac{\log(T/R_1)}{\log(1 + a_Q)} < \infty.$$

This proves the claim. Choose a subsequence of the recorded maps whose areas converge to $E \leq C - k\hbar_X$. The open-disk assertion of Lemma 2.8 now gives either a required sphere or a normalized principal map on $\mathbb{D}_{R_*}$ of area at most $E - \hbar_X \leq C - (k+1)\hbar_X$. If $R_* = T$, retain that disk; otherwise this is the remaining alternative in P_{k+1} . The repeated calls to P_k do not increase the nesting depth: only this last principal recovery adds one level.

Hence P_k holds for every $k \geq 1$. With $k = \lfloor C/\hbar_X \rfloor + 1$, its remaining principal-disk alternative would have negative area and is impossible. A required sphere or disk must therefore occur. The construction may use countably many finite stages and subsequence selections, while its recovery depth has the stated finite bound. $\square$

Proof of Theorem 1.1. Choose $F \Subset V \Subset N$ as above and put $\rho = 1 + \sup_{\overline{V}} |\eta \circ \psi|$. Every normalized disk f of radius greater than ρ must leave V before source radius ρ . Indeed, suppose that $f(\mathbb{D}_\rho) \subset V$

and set $b = \eta \circ \psi \circ f$. Then b is holomorphic on $\mathbb{D}_\rho$, while normalization gives $b(0) = 0$ and $b'(0) = 1$. Moreover,

$$\sup_{\mathbb{D}_\rho} |b| \leq \sup_{\overline{V}} |\eta \circ \psi| = \rho - 1.$$

For every $0 < r < \rho$, Cauchy's estimate for the derivative at the origin therefore gives

$$1 = |b'(0)| \leq \frac{1}{r} \sup_{|z|=r} |b(z)| \leq \frac{\rho - 1}{r}.$$

Letting $r \uparrow \rho$ yields $1 \leq (\rho - 1)/\rho < 1$, a contradiction. Hence there is $\zeta \in \mathbb{D}_\rho$ with $f(\zeta) \notin V$. Since $f(0) \in F \subset V$, continuity gives a first exit along the segment $t\zeta$, $0 \leq t \leq 1$: if

$$t_0 = \sup\{t \in [0, 1] : f(s\zeta) \in V \text{ for every } 0 \leq s < t\},$$

then $z = t_0\zeta$ satisfies $0 < |z| < \rho$ and $f(z) \in \partial V$.

For each $j \geq 2$, set $\delta_j = 1/(2j)$ and $C_j = A_* + 1/j$. Apply Proposition 2.10 with cap C_j and target radius $T_j = 2j\rho$, starting from the same seed, whose area is less than A_* . If it returns a disk h_j of radius at least T_j , choose its first exit z_j as above and set $g_j(\zeta) = h_j(z_j\zeta)$. This map is defined on a disk of radius greater than $2j$, so its restriction to $\mathbb{D}_j$ extends past $\mathbb{D}_j$ even when h_j is defined only on its open disk. If instead the hierarchy returns a sphere u , then $u(\mathbb{P}^1)$ meets ∂V . Indeed, if $u(\mathbb{P}^1) \subset V$, then $\psi \circ u : \mathbb{P}^1 \rightarrow \mathbb{C}^{d_Y}$ is holomorphic and hence constant. Since $u(\mathbb{P}^1)$ meets $F = \psi^{-1}(0)$, this constant is zero, so $u(\mathbb{P}^1) \subset F$, contrary to the conclusion of the hierarchy. Choose an affine coordinate on the sphere in which selected preimages of F and ∂V are 0 and 1, and restrict to $\mathbb{D}_j$ to define g_j . In both cases,

$$g_j(0) \in F, \quad g_j(1) \in \partial V, \quad \text{Area}_j(g_j) \leq A_* + 1/j.$$

Lemma 2.5 gives a chain between these target sets of total area at most A_* . Its first component not contained in F meets F at the first marked value or at the preceding node, and is the required sphere. $\square$

Proof of Theorem 1.2. Given $x \in X$, let N be a coordinate neighborhood of x and let $q : N \rightarrow Y \subset \mathbb{C}^n$ be its coordinate map. The fiber over $q(x)$ is $F = \{x\}$, and q is a submersion. Theorem 1.1 therefore gives a nonconstant sphere through x with area at most A_* . $\square$

Remark 2.11. On $\mathbb{P}^n$, use the Fubini–Study form

$$\omega_{\text{FS}} = \frac{i}{2} \partial \bar{\partial} \log(1 + |w^1|^2 + \cdots + |w^n|^2).$$

In the normalization (1), $\text{Ric}(\omega_{\text{FS}}) = 2(n+1)\omega_{\text{FS}}$ and $\int_{\mathbb{P}^1} \omega_{\text{FS}} = \pi$ on a projective line. Thus, with $\kappa = 2(n+1)$, the bound in Theorem 1.1 becomes π . Every nonconstant sphere in $\mathbb{P}^n$ has positive integer degree and hence area at least π , so this value cannot be decreased.

3. RICCI TRACES AND SCALAR-CONSTRAINED VARIATIONS

We now use positive Ricci curvature to force area decrease above A_* . The calculation combines a boundary trace identity with a choice of holomorphic multipliers that fix the center and one scalar derivative. Their Dirichlet cost $\pi(n+1)$, together with Proposition 3.4, gives the threshold $A_* = 2\pi(n+1)/\kappa$. Section 4 realizes the resulting variation fields by holomorphic families with uniform estimates. We retain the normalizations (1)–(2).

For $z = x + iy$, the complex derivative and the Euclidean Laplacian are

$$\partial_z = \frac{1}{2}(\partial_x - i\partial_y), \quad \Delta = \partial_x^2 + \partial_y^2 = 4\partial_z\partial_{\bar{z}}.$$

On f^*T_X , ∇_z denotes the Chern covariant derivative induced by ω in the source direction. On $\partial\mathbb{D}_R$, ν is the outward Euclidean unit normal, ∂_ν the corresponding derivative, and ds Euclidean

arclength. Let A^* denote the conjugate transpose of A , and let Id_n denote the identity matrix. We use the Hilbert–Schmidt norm and the operator norm

$$\|A\|_{\text{HS}}^2 := \text{tr}(AA^*) = \sum_{j,k} |A_{jk}|^2, \quad \|A\|_{\text{op}} := \sup_{|v|=1} |Av|. \quad (21)$$

Distances and diameters are measured with the Kähler metric ω .

For a Hermitian holomorphic line bundle (L, h) , let $\Theta(L)$ denote the curvature of its Chern connection. We use the real curvature form $i\Theta(L) = -i\partial\bar{\partial}\log|s|_h^2$, where s is a nonvanishing local holomorphic frame, and its cohomology class is $2\pi c_1(L)$ [9, Ch. V, (12.8), (13.3)]. We write $c_1(X) = c_1(K_X^{-1})$.

The following proposition is due to Yau [50]; it is an immediate consequence of his prescribed-Ricci theorem. For the reader's convenience, we include a proof.

Proposition 3.1 (Yau). *Every complex Fano manifold admits a Kähler metric ω and a constant $\kappa > 0$ such that*

$$\text{Ric}(\omega) \geq \kappa\omega. \quad (22)$$

Proof. Ampleness of K_X^{-1} gives a positive closed $(1, 1)$ -form ρ representing $2\pi c_1(X)$. In a fixed Kähler class, Yau's prescribed-Ricci theorem gives a metric ω with $\text{Ric}(\omega) = \rho$. The least eigenvalue of ρ relative to ω is positive and continuous on the compact manifold. Its positive minimum therefore supplies κ with $\text{Ric}(\omega) \geq \kappa\omega$. $\square$

The size of κ depends on the scale: $\text{Ric}(c\omega) = \text{Ric}(\omega)$ for $c > 0$, so rescaling replaces κ by κ/c . In the class $[\omega] = 2\pi c_1(X)$, Tian's anticanonical α -invariant gives a quantitative choice: for every $0 < \kappa < \min\{1, (n+1)\alpha(X)/n\}$, one can choose ω satisfying (22); see [45] and [44, p. 324]. This concerns the choice of metric; our construction works with any fixed metric satisfying the stated Ricci bound.

Remark 3.2. On a Fano manifold, (22) specifies the metric used in the construction. The form $\text{Ric}(\omega)$ is the Chern curvature of the induced metric on K_X^{-1} . Its positivity expresses the ampleness of K_X^{-1} in differential-geometric terms, in accordance with Kodaira's embedding theorem [30, pp. 314–316].

In Euclidean space, the paired second area variation is Green's formula for the squared norm of the variation field. The *Kähler condition* $d\omega = 0$ makes the same boundary identity exact for a curved target, by the classical coefficient symmetries of closed $(1, 1)$ -forms [9, Ch. VI, §4].

Lemma 3.3. *Let X be a complex manifold equipped with a Kähler metric ω , and fix $R > 0$. Let $\Delta_t \subset \mathbb{C}$ be a parameter disk centered at zero, $U \subset \mathbb{C}$ an open neighborhood of $\mathbb{D}_R$, and $\mathcal{F} : U \times \Delta_t \rightarrow X$ a holomorphic map. Write $f_t(z) := \mathcal{F}(z, t)$ and $\xi := \partial_t \mathcal{F}(\cdot, 0)$. Denote the second derivatives along the two real parameter axes by*

$$H_{\mathbb{R}} := \left. \frac{d^2}{ds^2} \right|_{s=0} \text{Area}_R(f_s), \quad H_{i\mathbb{R}} := \left. \frac{d^2}{ds^2} \right|_{s=0} \text{Area}_R(f_{is}).$$

Then

$$H_{\mathbb{R}} + H_{i\mathbb{R}} = \int_{\partial\mathbb{D}_R} \partial_\nu |\xi|_\omega^2 ds. \quad (23)$$

Two holomorphic families with the same central map f_0 and the same tangent field ξ therefore have the same paired second variation $H_{\mathbb{R}} + H_{i\mathbb{R}}$.

Proof. Set $\tilde{\omega} := \mathcal{F}^*\omega$, and write its Hermitian coefficient matrix in the source coordinates $\zeta^1 = z$, $\zeta^2 = t$ as

$$\tilde{\omega} = \frac{i}{2} \sum_{\alpha, \beta=1}^2 \tilde{\omega}_{\alpha\bar{\beta}} d\zeta^\alpha \wedge d\bar{\zeta}^\beta.$$

The Kähler condition gives $d\tilde{\omega} = \mathcal{F}^*(d\omega) = 0$. Closedness relates derivatives in the parameter direction to derivatives in the source direction. Comparing the $(2, 1)$ and $(1, 2)$ coefficients yields

$$\partial_t \tilde{\omega}_{z\bar{z}} = \partial_z \tilde{\omega}_{t\bar{z}}, \quad \partial_{\bar{t}} \tilde{\omega}_{t\bar{z}} = \partial_{\bar{z}} \tilde{\omega}_{t\bar{t}}.$$

Differentiating these identities and commuting mixed derivatives gives

$$\partial_t \partial_{\bar{t}} \tilde{\omega}_{z\bar{z}} = \partial_z \partial_{\bar{t}} \tilde{\omega}_{t\bar{z}} = \partial_z \partial_{\bar{z}} \tilde{\omega}_{t\bar{t}}.$$

At $t = 0$, the coefficient $\tilde{\omega}_{t\bar{t}}$ is $|\xi|_\omega^2$. Consequently

$$\partial_t \partial_{\bar{t}} \text{Area}_R(f_t)|_{t=0} = \int_{\mathbb{D}_R} \partial_z \partial_{\bar{z}} |\xi|_\omega^2 dA = \frac{1}{4} \int_{\partial \mathbb{D}_R} \partial_\nu |\xi|_\omega^2 ds.$$

The sum of the second derivatives along the real and imaginary parameter axes is the parameter Laplacian $4\partial_t \partial_{\bar{t}}$. Hence

$$H_{\mathbb{R}} + H_{i\mathbb{R}} = 4 \partial_t \partial_{\bar{t}} \text{Area}_R(f_t)|_{t=0},$$

which yields (23). $\square$

The classical Riemannian second-variation formula contains interior curvature and second-fundamental-form terms; on Kähler submanifolds, holomorphic normal fields are Jacobi fields [41, §§3.2, 3.5]. In our families the boundary values are allowed to move, so the first variation of area need not vanish. Nevertheless, holomorphicity and $d\omega = 0$ give the exact paired boundary formula (23); the calculation also remains valid at branch points.

Fix $R > 0$ and a map f holomorphic on a neighborhood of $\overline{\mathbb{D}_R}$. To select variation fields before realizing them by holomorphic families, define the boundary quadratic form on holomorphic sections ξ of f^*T_X that are smooth on the closed disk:

$$\mathcal{H}_{R,f}(\xi) := \int_{\partial \mathbb{D}_R} \partial_\nu |\xi|_\omega^2 ds. \quad (24)$$

When ξ is the tangent field of a complex holomorphic family, Lemma 3.3 identifies $\mathcal{H}_{R,f}(\xi)$ with the sum of the second area derivatives along the real and imaginary parameter axes.

We use the Wiener–Masani boundary normalization in the form employed by Berndtsson and Rosay [3, Theorem 2.1 and p. 888]; Lemma 4.1 recalls the construction of the resulting boundary-unitary holomorphic frame. The following computation combines this normalization with Green’s formula and the classical determinant-curvature identities [9, Ch. V, (13.3); Ch. VIII, (6.6)–(6.7)]. We record the matrix form that separates the Ricci integral from the multiplier’s Dirichlet cost and will give the descent threshold below.

Proposition 3.4. *Let (X, ω) be a Hermitian complex n -fold, let $R > 0$, and let $f : U \rightarrow X$ be holomorphic on an open neighborhood $U \supset \overline{\mathbb{D}_R}$. Let $e = (e_1, \dots, e_n)$ be a holomorphic frame of f^*T_X , smooth on the closed disk and unitary on $\partial \mathbb{D}_R$. For a positive integer N , let Φ be a holomorphic $\text{Mat}_{n \times N}(\mathbb{C})$ -valued map on a neighborhood of $\overline{\mathbb{D}_R}$, satisfying the boundary coisometry condition*

$$\Phi(\zeta) \Phi(\zeta)^* = \text{Id}_n \quad (|\zeta| = R).$$

The columns Φ_a give sections $(e\Phi)_a = \sum_{j=1}^n e_j \Phi_{ja}$ satisfying the trace identity

$$\sum_{a=1}^N \mathcal{H}_{R,f}((e\Phi)_a) = 4 \int_{\mathbb{D}_R} \|\Phi'\|_{\text{HS}}^2 dA - 2 \int_{\mathbb{D}_R} f^* \text{Ric}(\omega). \quad (25)$$

Here Φ' denotes the entrywise derivative, and $\|\cdot\|_{\text{HS}}$ is the norm in (21).

Proof. The Gram matrix $G := (\langle e_j, e_k \rangle_\omega)_{j,k}$ represents squared norms by $|ec|_\omega^2 = c^\top G \bar{c}$. Thus

$$\sum_{a=1}^N |(e\Phi)_a|_\omega^2 = \sum_{a=1}^N \Phi_a^\top G \bar{\Phi}_a = \text{tr}(\Phi^\top G \bar{\Phi}) = \text{tr}(G \bar{\Phi} \Phi^\top).$$

The first trace sums the diagonal entries of $\Phi^\top G \bar{\Phi}$; the last equality uses cyclicity of the trace. Since finite summation commutes with the normal derivative and integration, (24) gives the boundary integral of $\partial_\nu \text{tr}(G \bar{\Phi} \Phi^\top)$. On the boundary, $G = \bar{\Phi} \Phi^\top = \text{Id}_n$, but their normal derivatives need not vanish. The product rule and Jacobi's classical determinant formula (see, for instance, [4]) give, on $\partial \mathbb{D}_R$,

$$\begin{aligned}\partial_\nu \text{tr}(G \bar{\Phi} \Phi^\top) &= \text{tr}(\partial_\nu G) + \text{tr}(\partial_\nu (\bar{\Phi} \Phi^\top)), \\ \partial_\nu \log \det G &= \text{tr}(G^{-1} \partial_\nu G) = \text{tr}(\partial_\nu G).\end{aligned}$$

The boundary normalization is essential in the second line: tr and $\log \det$ have the same first differential at Id_n . The divergence theorem and $\Delta \text{tr}(\bar{\Phi} \Phi^\top) = 4 \|\Phi'\|_{\text{HS}}^2$ give the multiplier term. Since $\det G$ is the metric coefficient on $f^* K_X^{-1}$, (1) gives

$$f^* \text{Ric}(\omega) = -\frac{1}{2} \Delta \log \det G dA.$$

Green's formula therefore turns the determinant term $\partial_\nu \log \det G$ into $-2 \int f^* \text{Ric}(\omega)$. $\square$

We first construct variations that fix the value of a disk at 0 and one nonzero scalar component of its derivative. Fix $x_0 \in X$ and a nonzero complex-linear covector

$$\lambda \in T_{x_0}^* X \setminus \{0\}. \quad (26)$$

The scalar condition $\lambda(f_z(0)) = 1$ fixes one component of the initial derivative and gives $|f_z(0)|_\omega \geq \|\lambda\|_{\omega,*}^{-1}$, where the norm on the covector is dual to ω at x_0 . The multiplier below imposes the linearized value and scalar-derivative constraints. Proposition 2.1 realizes these constraints by holomorphic families that preserve $f(0) = x_0$ and $\lambda(f_z(0)) = 1$ exactly, while leaving the other derivative components free. In Section 2, the center varies in a compact fiber and the covector is pulled back from the base. In the principal branch of Lemma 2.6, this base normalization passes to the limit. For a disk satisfying $f(0) = x_0$ and $\lambda(f_z(0)) = 1$, the frame value $e(0) : \mathbb{C}^n \rightarrow T_{x_0} X$ identifies the scalar constraint with the nonzero covector $L_f := \lambda \circ e(0)$. Choose $U_f \in U(n)$ so that its last $n-1$ columns form an orthonormal basis of $\ker L_f$. Fixing the center requires every column of the multiplier to vanish at zero; in the last $n-1$ directions, which lie in $\ker L_f$, a simple zero also preserves the scalar derivative to first order, while in the remaining direction a double zero makes its derivative at zero vanish. The resulting multiplier is

$$\Phi_f(z) := U_f \text{diag} \left((z/R)^2, z/R, \dots, z/R \right). \quad (27)$$

The columns of Φ_f vanish at zero, so $(e\Phi_f)'(0) = e(0)\Phi_f'(0)$. The first column of Φ_f has zero derivative at zero; the derivatives of its remaining columns belong to $\ker L_f \subset \mathbb{C}^n$. Thus $\Phi_f(0) = 0$ and $L_f \Phi_f'(0) = 0$ impose the infinitesimal constraints. On $|z| = R$, the diagonal entries have modulus one, so $\Phi_f \Phi_f^* = \text{Id}_n$. Left multiplication by the unitary matrix U_f preserves the Hilbert–Schmidt norm. Differentiating the diagonal entries therefore gives

$$\left\| \Phi_f'(z) \right\|_{\text{HS}}^2 = \frac{4|z|^2}{R^4} + \frac{n-1}{R^2}.$$

Integrating in polar coordinates, we obtain

$$\int_{\mathbb{D}_R} \left\| \Phi_f' \right\|_{\text{HS}}^2 dA = 2\pi \int_0^R \left(\frac{4r^2}{R^4} + \frac{n-1}{R^2} \right) r dr = \pi(n+1). \quad (28)$$

The Ricci lower bound for our chosen metric gives

$$\sum_{a=1}^n \mathcal{H}_{R,f}((e\Phi_f)_a) \leq 4\pi(n+1) - 2\kappa \text{Area}_R(f). \quad (29)$$

The cost in (28) is optimal within this multiplier class. For a nonzero covector $L \in (\mathbb{C}^n)^*$, suppose that Φ is holomorphic near $\overline{\mathbb{D}}_R$, takes values in $\text{Mat}_{n \times N}(\mathbb{C})$, and satisfies

$$\Phi\Phi^* = \text{Id}_n \quad \text{on } \partial\mathbb{D}_R, \quad \Phi(0) = 0, \quad L\Phi'(0) = 0.$$

Choose $Q \in U(n)$ with first row $L/\|L\|$. Replacing Φ by $Q\Phi$ preserves the boundary identity and the Hilbert–Schmidt Dirichlet integral. Its first row vanishes to order at least two, while the remaining rows vanish to order at least one. Each row has norm one on $\partial\mathbb{D}_R$. For any such row $r(z) = \sum_{k \geq m} a_k(z/R)^k$, Parseval’s identity and radial integration give

$$\sum_{k \geq m} \|a_k\|^2 = 1, \quad \int_{\mathbb{D}_R} |r'|^2 dA = \pi \sum_{k \geq m} k \|a_k\|^2 \geq \pi m.$$

Summing over the rows gives $2\pi + (n-1)\pi = \pi(n+1)$, attained by the multiplier in (27).

4. UNIFORM HOLOMORPHIC REALIZATION AND DISK OPERATIONS

We realize the trace calculation by holomorphic families and prove the quantitative disk operations stated in Propositions 2.1 and 2.2. Holomorphic mapping spaces and Stein neighborhoods of graphs provide a qualitative deformation theory [15]. Both the parameter radius and the cubic area remainder must remain uniform as the disk and its source radius vary. This control is obtained by lifting to a fixed Stein submanifold of affine space, perturbing there, and projecting back to X .

Assume in this section that X is projective. Keep the projective embedding $\iota : X \hookrightarrow \mathbb{P}^{m-1}$ fixed as in Section 2; it will be used to represent $\iota \circ f$ by homogeneous coordinates and to construct the lift space $\mathcal{Y}$. The embedding ι is kept fixed as f and R vary, so the constants below may depend on ι .

We use the standard L^p and C^k norms on source regions, with Euclidean area measure. Vector and operator norms are induced by the fixed Hermitian and Euclidean metrics; matrix-valued L^2 estimates use the Hilbert–Schmidt norm.

4.1. Frame and lifting estimates

Bounds on area and weighted derivative do not control derivatives on the source boundary. The following example shows that uniform realization requires direct estimates. For $0 < r < 1$ and $\varepsilon > 0$, the functions $h_r : \mathbb{D} \rightarrow \mathbb{C}$ given by

$$h_r(z) := \varepsilon \frac{1-r^2}{1-rz}$$

are holomorphic near $\overline{\mathbb{D}}$. A direct computation gives

$$\int_{\mathbb{D}} |h'_r|^2 dA = \pi \varepsilon^2 r^2, \quad B_1(h_r) \leq \frac{\varepsilon}{2}, \quad |h'_r(1)| = \varepsilon \frac{r(1+r)}{1-r} \rightarrow \infty.$$

Bounded area and bounded B_1 therefore do not imply relative compactness in $C^1(\overline{\mathbb{D}})$ (with the Euclidean metric on $\mathbb{C}$); the uniform estimates below must be proved directly.

Boundary spectral factorization and a Green potential bound control a normalized frame. Homogeneous coordinates and Nicolau’s corona theorem [37] then give a uniformly bounded affine lift with bounded Dirichlet integral.

Fix $A_0, B_0 \geq 0$. In the auxiliary results below, f is holomorphic near $\overline{\mathbb{D}}_R$ and satisfies $\text{Area}_R(f) \leq A_0$ and $B_R(f) \leq B_0$. Norms on fixed finite-rank bundles are chosen once and for all.

The spectral factorization theorem is due to Wiener and Masani [46, Theorem 7.13]. We use the smooth boundary version stated by Berndtsson and Rosay [3, Theorem 2.1]. It makes the frame in Theorem 4.7 unitary on the boundary, as required by the Ricci-trace identity.

Lemma 4.1. *Let $H : \partial\mathbb{D} \rightarrow \text{Mat}_{n \times n}(\mathbb{C})$ be a smooth positive-definite Hermitian matrix function. There is a map $A : \mathbb{D} \rightarrow \text{GL}_n(\mathbb{C})$, holomorphic in $\mathbb{D}$ and smooth on $\overline{\mathbb{D}}$ with invertible boundary values, such that*

$$A(\zeta)^* H(\zeta) A(\zeta) = \text{Id}_n \quad (|\zeta| = 1).$$

Proof. The Wiener–Masani theorem in the smooth boundary form of [3, Theorem 2.1] gives $H = h^* h$, where h and h^{-1} are holomorphic on $\mathbb{D}$ and smooth on $\overline{\mathbb{D}}$. Thus $A := h^{-1}$ has the required boundary normalization. $\square$

For $z \in \mathbb{D}$, let δ_z be the unit point mass at z . The Dirichlet Green function of the unit disk, with the sign and normalization used here, is [9, Ch. I, §4.B]

$$g(z, w) = \log \left| \frac{1 - \bar{z}w}{z - w} \right|, \quad -\Delta_w g = 2\pi\delta_z.$$

On $\mathbb{D}_R$ we use $g_R(z, w) := g(z/R, w/R)$.

The Green potential controls both the frame and the homogeneous lift. The following elementary estimate separates a neighborhood of the logarithmic singularity, controlled by B_0 , from its complement, controlled by the total area.

Lemma 4.2. *Let (X, ω) be a Hermitian complex manifold, let $R > 0$ and $A_0, B_0 \geq 0$, and let $f : U \rightarrow X$ be holomorphic on an open neighborhood $U \supset \overline{\mathbb{D}}_R$. Let $\varepsilon_f^\omega(w) := |f_z(w)|_\omega^2$ be the area density of f , so that $f^*\omega = \varepsilon_f^\omega dA$ and $\text{Area}_R(f) = \int_{\mathbb{D}_R} \varepsilon_f^\omega dA$. If $\text{Area}_R(f) \leq A_0$ and $B_R(f) \leq B_0$, then*

$$\sup_{z \in \mathbb{D}_R} \int_{\mathbb{D}_R} g_R(z, w) \varepsilon_f^\omega(w) dA(w) \leq C_g(A_0, B_0) := C_G(B_0^2 + A_0), \quad (30)$$

where C_G is a universal constant, independent of R and f .

Proof. First reduce to the unit disk. Under the dilation $h(\zeta) := f(R\zeta)$, the identities

$$\text{Area}_1(h) = \text{Area}_R(f), \quad B_1(h) = B_R(f), \quad \varepsilon_h^\omega(\eta) = R^2 \varepsilon_f^\omega(R\eta)$$

show that the hypotheses are unchanged. Since $g_R(R\zeta, R\eta) = g(\zeta, \eta)$, change of variables also preserves the Green-potential integral. We may therefore assume $R = 1$.

Fix $z \in \mathbb{D}$ and set $\delta := 1 - |z|$, its distance to the boundary. Split the integral into the disk $|w - z| < \delta/2$ and its complement in $\mathbb{D}$. On the first region,

$$1 - |w| \geq 1 - |z| - |w - z| > \frac{\delta}{2}, \quad |f_z(w)|_\omega \leq \frac{B_0}{1 - |w|}.$$

Thus $\varepsilon_f^\omega(w) \leq 4B_0^2/\delta^2$. To bound the Green kernel, write

$$1 - \bar{z}w = (1 - |z|^2) + \bar{z}(z - w).$$

Since $1 - |z|^2 = (1 + |z|)\delta \leq 2\delta$, the triangle inequality gives

$$|1 - \bar{z}w| \leq 2\delta + |z - w| \leq \frac{5\delta}{2}, \quad g(z, w) \leq \log \frac{5\delta}{2|z - w|}.$$

Integration in polar coordinates bounds this contribution by

$$\frac{4B_0^2}{\delta^2} 2\pi \int_0^{\delta/2} r \log \frac{5\delta}{2r} dr = \pi B_0^2 \left(\log 5 + \frac{1}{2} \right).$$

On the complementary region $|z - w| \geq \delta/2$, the same triangle inequality instead yields

$$\frac{|1 - \bar{z}w|}{|z - w|} \leq 1 + \frac{2\delta}{|z - w|} \leq 5.$$

Thus $g(z, w) \leq \log 5$ there, and the remaining contribution is at most

$$(\log 5) \int_{\mathbb{D}} \varepsilon_f^\omega dA \leq A_0 \log 5.$$

Adding the two bounds proves (30); one may take the universal constant $C_G := \pi(\log 5 + 1/2)$, independently of z , R , and f . $\square$

We use the following Chern connection and curvature convention [9, Ch. V, (12.2)–(12.4)]. In the coordinates of (1), a section with coefficient column c has squared norm $c^\top h \bar{c}$. Writing $\Theta(T_X) = \nabla^2$, the connection and curvature matrices are

$$\nabla' c = \partial c + \bar{h}^{-1}(\partial \bar{h})c, \quad \nabla'' c = \bar{\partial} c, \quad \Theta(T_X) = \bar{\partial}(\bar{h}^{-1} \partial \bar{h}).$$

The coefficient of $dw^\alpha \wedge d\bar{w}^\beta$ in $\Theta(T_X)$ is $-\partial_{\bar{w}^\beta}(\bar{h}^{-1} \partial_{w^\alpha} \bar{h})$; the minus sign comes from reversing the order of the differentials. For $v = \sum_\alpha v^\alpha \partial_{w^\alpha}$ and a tangent vector u with coefficient column c , our curvature tensor is

$$R^\omega(v, \bar{v}, u, \bar{u}) := \sum_{\alpha, \beta} v^\alpha \bar{v}^\beta c^\top [-\partial_{w^\alpha} \partial_{\bar{w}^\beta} h + (\partial_{w^\alpha} h) h^{-1} (\partial_{\bar{w}^\beta} h)] \bar{c}. \quad (31)$$

Positive holomorphic bisectional curvature therefore means that (31) is positive for every pair of nonzero tangent vectors v, u at the same point. For a holomorphic disk f , the curvature endomorphism $\mathcal{R}_f^\omega$ is characterized by

$$\langle \mathcal{R}_f^\omega v, v \rangle_\omega = R^\omega(f_z, \bar{f}_z, v, \bar{v}). \quad (32)$$

The Bochner formula and the curvature formula for the determinant bundle [9, Ch. VIII, proof of Lemma 7.2; Ch. V, (4.6)] give, for a holomorphic section s of f^*T_X ,

$$\partial_z \partial_{\bar{z}} |s|_\omega^2 = |\nabla_z s|_\omega^2 - \langle \mathcal{R}_f^\omega s, s \rangle_\omega, \quad f^* \text{Ric}(\omega) = 2 \text{tr}(\mathcal{R}_f^\omega) dA. \quad (33)$$

Set

$$C_\omega := \max_{x \in X} \sup_{\substack{\xi, u \in T_x X \\ |\xi|_\omega = |u|_\omega = 1}} |R^\omega(\xi, \bar{\xi}, u, \bar{u})| < \infty.$$

Then

$$\|\mathcal{R}_f^\omega(z)\|_{\text{op}} \leq C_\omega |f_z(z)|_\omega^2 = C_\omega \varepsilon_f^\omega(z).$$

For a frame $e = (e_1, \dots, e_n)$, write $G = (\langle e_j, e_k \rangle_\omega)$. Matrix inequalities refer to Hermitian quadratic forms. The Green bound and Bochner identity control G throughout the disk and bound the frame's covariant Dirichlet integral. Both estimates enter the lifted-field bounds in Theorem 4.7. For the two-sided metric control we apply Berndtsson and Rosay's curvature-potential estimate [3, Proposition 2.3], making the dependence on A_0, B_0 explicit. The covariant Dirichlet bound then follows from the integrated Bochner identity.

Proposition 4.3. *Let (X, ω) be a compact Hermitian complex n -fold, and fix $A_0, B_0 \geq 0$. There are constants $\Gamma \geq 1$ and $D_1 < \infty$, depending only on (X, ω, A_0, B_0) , with the following property. For any $R > 0$ and any f holomorphic on a neighborhood of $\mathbb{D}_R$ with $\text{Area}_R(f) \leq A_0$ and $B_R(f) \leq B_0$, every holomorphic frame e of f^*T_X that is smooth on the closed disk and unitary on its boundary has Gram matrix G satisfying*

$$\Gamma^{-1} \text{Id}_n \leq G \leq \Gamma \text{Id}_n. \quad (34)$$

Moreover

$$\sum_{a=1}^n \int_{\mathbb{D}_R} |\nabla_z e_a|^2 dA \leq D_1. \quad (35)$$

Both Γ and D_1 are independent of R , f , and the choice of boundary-unitary frame.

Proof. Set

$$P(z) = \int_{\mathbb{D}_R} g_R(z, w) \varepsilon_f^\omega(w) dA(w), \quad \phi(z) = \frac{C_\omega}{\pi} C_g(A_0, B_0) - \frac{2C_\omega}{\pi} P(z).$$

The Green normalization and Lemma 4.2 give

$$\phi_{z\bar{z}} = C_\omega \varepsilon_f^\omega, \quad \|\phi\|_\infty \leq \frac{C_\omega}{\pi} C_g(A_0, B_0).$$

In the holomorphic frame e , the metric is the identity on the boundary, and its curvature satisfies $|\langle \mathcal{R}_f^\omega v, v \rangle_\omega| \leq \phi_{z\bar{z}} |v|_\omega^2$. Berndtsson–Rosay’s estimate [3, Proposition 2.3], applied after dilation to the unit disk, therefore bounds the metric between $e^{-2\|\phi\|_\infty} \text{Id}_n$ and $e^{2\|\phi\|_\infty} \text{Id}_n$. Both $\text{tr } G$ and $\text{tr}(G^{-1})$ are at most

$$\Gamma := n \exp\left(\frac{2C_\omega C_g(A_0, B_0)}{\pi}\right),$$

which gives (34).

It remains to bound the frame’s covariant Dirichlet integral. We first sum the Bochner identity; after integration, the boundary determinant calculation from Proposition 3.4 will identify its left-hand side:

$$\partial_z \partial_{\bar{z}} \text{tr } G = \sum_a |\nabla_z e_a|^2 - \text{tr}(\mathcal{R}_f^\omega \mathbf{S}), \quad \mathbf{S} := \sum_a e_a \otimes e_a^b.$$

Here $e_a^b(v) = \langle v, e_a \rangle_\omega$ denotes metric duality. Since $G = \text{Id}_n$ on $\partial \mathbb{D}_R$, Jacobi’s formula gives

$$\partial_\nu \log \det G = \text{tr}(G^{-1} \partial_\nu G) = \partial_\nu \text{tr } G \quad \text{on } \partial \mathbb{D}_R.$$

Consequently Green’s formula and (1) give

$$\int_{\mathbb{D}_R} \partial_z \partial_{\bar{z}} \text{tr } G dA = \frac{1}{4} \int_{\partial \mathbb{D}_R} \partial_\nu \log \det G ds = -\frac{1}{2} \int_{\mathbb{D}_R} f^* \text{Ric}(\omega).$$

Integrating the Bochner identity and solving for the nonnegative derivative term therefore gives

$$\sum_a \int_{\mathbb{D}_R} |\nabla_z e_a|^2 dA = \int_{\mathbb{D}_R} \text{tr}(\mathcal{R}_f^\omega \mathbf{S}) dA - \frac{1}{2} \int_{\mathbb{D}_R} f^* \text{Ric}(\omega).$$

Now $\text{tr } \mathbf{S} \leq n\Gamma$, while the absolute Ricci density is at most $2nC_\omega \varepsilon_f^\omega$, since $f^* \text{Ric} = 2 \text{tr}(\mathcal{R}_f^\omega) dA$. Taking absolute values and using $\text{Area}_R(f) \leq A_0$ gives

$$\sum_a \int_{\mathbb{D}_R} |\nabla_z e_a|^2 dA \leq nC_\omega (\Gamma + 1) A_0.$$

$D_1 := nC_\omega (\Gamma + 1) A_0$ is therefore an admissible choice in (35). $\square$

Recall the projective embedding $\iota : X \hookrightarrow \mathbb{P}^{m-1}$ chosen at the beginning of this section. Let ω_{FS} be the Fubini–Study form on its target, with normalization

$$\omega_{\text{FS}} = \frac{i}{2} \partial \bar{\partial} \log \sum_{j=1}^m |Z_j|^2.$$

Choose $c_\iota > 0$ such that $\iota^* \omega_{\text{FS}} \leq c_\iota \omega$; such a constant exists by compactness of X . The displayed Fubini–Study formula is read in homogeneous coordinates $Z = (Z_1, \dots, Z_m) \neq 0$; it is unchanged as a form on projective space when the homogeneous representative is multiplied by a nonvanishing holomorphic function.

In rank one, boundary normalization subtracts a harmonic function from the logarithm of the metric, as in Berndtsson and Rosay [3, p. 886]. Together with the classical Fubini–Study potential [9, Ch. VI, (4.4)], this gives homogeneous coordinates with the bounds needed for the corona theorem. We record the area identity and the uniform lower bound.

Lemma 4.4. *Let (X, ω) be a compact projective manifold with a smooth Hermitian metric, and fix an embedding $\iota : X \hookrightarrow \mathbb{P}^{m-1}$ and bounds $A_0, B_0 \geq 0$. For every $R > 0$, a map $f : \mathbb{D}_R \rightarrow X$ holomorphic on a neighborhood of $\overline{\mathbb{D}}_R$ with $\text{Area}_R(f) \leq A_0$ and $B_R(f) \leq B_0$ has a homogeneous lift $\mathbf{s}$ from an open neighborhood of $\overline{\mathbb{D}}_R$ to $\mathbb{C}^m \setminus \{0\}$, holomorphic on that neighborhood, with $[\mathbf{s}] = \iota \circ f$ and $|\mathbf{s}| = 1$ on the boundary. It satisfies*

$$|\mathbf{s}| \leq 1, \quad \int_{\mathbb{D}_R} |\mathbf{s}'|^2 dA = \int_{\mathbb{D}_R} f^* \iota^* \omega_{\text{FS}} \leq c_\iota A_0, \quad (36)$$

and

$$|\mathbf{s}(z)| \geq \delta_{\mathbf{s}} > 0, \quad (37)$$

where $\delta_{\mathbf{s}} = \delta_{\mathbf{s}}(X, \omega, \iota, A_0, B_0)$ is independent of R and f .

Proof. The tautological line bundle on $\mathbb{P}^{m-1}$ has fiber $\mathbb{C}Z$ over $[Z]$. Trivialize its pullback by $\iota \circ f$ on a slightly larger disk, using the Oka–Grauert principle [20, Satz 6], and choose a nonvanishing section, represented by a lift v in $\mathbb{C}^m \setminus \{0\}$. The positive real-analytic boundary function $|v|$ has a logarithm whose harmonic extension is the real part of a holomorphic function a on a slightly larger disk. Then $\mathbf{s} := e^{-a}v$ has boundary norm one, and the maximum principle gives $|\mathbf{s}| \leq 1$.

The area density of $\iota \circ f$ for the Fubini–Study metric is $\varepsilon_f^{\text{FS}} := |(\iota \circ f)'|_{\text{FS}}^2$, so that $f^* \iota^* \omega_{\text{FS}} = \varepsilon_f^{\text{FS}} dA$. Set $u := \log |\mathbf{s}|^2$. The boundary normalization $|\mathbf{s}| = 1$ gives $u = 0$ and $\partial_\nu u = \partial_\nu |\mathbf{s}|^2$ there. Since $\mathbf{s}$ has no common zero, the Fubini–Study formula gives $\Delta u = 4\varepsilon_f^{\text{FS}}$, while holomorphicity gives $\Delta |\mathbf{s}|^2 = 4|\mathbf{s}'|^2$. Green’s formula therefore yields

$$\int_{\mathbb{D}_R} |\mathbf{s}'|^2 dA = \frac{1}{4} \int_{\partial \mathbb{D}_R} \partial_\nu |\mathbf{s}|^2 ds = \frac{1}{4} \int_{\partial \mathbb{D}_R} \partial_\nu u ds = \int_{\mathbb{D}_R} \varepsilon_f^{\text{FS}} dA,$$

which proves (36). Green representation for u with zero boundary values [9, Ch. I, (4.5)] gives

$$-u(z) = \frac{2}{\pi} \int g_R(z, w) \varepsilon_f^{\text{FS}}(w) dA(w) \leq \frac{2c_\iota C_g(A_0, B_0)}{\pi}.$$

(37) therefore holds with $\delta_{\mathbf{s}} = \exp(-c_\iota C_g(A_0, B_0)/\pi)$. $\square$

Write $H^\infty(\mathbb{D})$ for the bounded holomorphic functions with the supremum norm, and let

$$\mathcal{D}(\mathbb{D}) := \left\{ h : \mathbb{D} \rightarrow \mathbb{C} \text{ holomorphic} : \int_{\mathbb{D}} |h'|^2 dA < \infty \right\}.$$

The Dirichlet algebra is

$$\mathfrak{A} := H^\infty(\mathbb{D}) \cap \mathcal{D}(\mathbb{D}), \quad \|h\|_{\mathfrak{A}} := \|h\|_\infty + \left(\int_{\mathbb{D}} |h'|^2 dA \right)^{1/2}.$$

To construct the affine lift, we need a holomorphic dual vector χ with $\chi \cdot \mathbf{s} = 1$ and bounds on both its supremum norm and Dirichlet integral. Nicolau’s theorem and the bound at the beginning of its proof [37, pp. 135–136] give the following statement.

Theorem 4.5. *For every $m \geq 1$ and $A, \delta > 0$ there is $C_N(m, A, \delta) < \infty$ such that, whenever $u_1, \dots, u_m \in \mathfrak{A}$ satisfy*

$$\|u_j\|_{\mathfrak{A}} \leq A, \quad \max_j |u_j(z)| \geq \delta \quad (z \in \mathbb{D}),$$

there are $v_1, \dots, v_m \in \mathfrak{A}$ with

$$\sum_j u_j v_j = 1, \quad \|v_j\|_{\mathfrak{A}} \leq C_N(m, A, \delta).$$

The same statement holds on every disk after dilation.

Proof. The case $\delta > A$ is vacuous; assume $\delta \leq A$. Nicolau uses the norm

$$\|h\|_N := \|h\|_\infty + \left(\frac{1}{\pi} \int_{\mathbb{D}} |h'|^2 dA \right)^{1/2}, \quad \|h\|_N \leq \|h\|_{\mathfrak{A}} \leq \sqrt{\pi} \|h\|_N.$$

His corona condition is the same maximum lower bound as above [37, p. 135, (1)]. The generators u_j/A therefore have Nicolau norm at most 1 and lower bound δ/A . The estimate at the beginning of his proof [37, p. 136] gives solutions b_j with $\sum_j (u_j/A) b_j = 1$ and $\|b_j\|_N \leq K_m(\delta/A)$, where K_m depends only on the indicated data. Setting $v_j := b_j/A$ gives $\sum_j u_j v_j = 1$ and

$$\|v_j\|_{\mathfrak{A}} \leq \frac{\sqrt{\pi}}{A} K_m(\delta/A),$$

which is an admissible choice of $C_N(m, A, \delta)$. For a disk of radius R ,

$$\|h(R \cdot)\|_{\infty, \mathbb{D}} = \|h\|_{\infty, \mathbb{D}_R}, \quad \int_{\mathbb{D}} |(h(R\zeta))'|^2 dA(\zeta) = \int_{\mathbb{D}_R} |h'(z)|^2 dA(z),$$

so the same bound applies after dilation. $\square$

For a normalized lift $\mathbf{s}$ from Lemma 4.4, its coordinates have algebra norm at most $C_s := 1 + \sqrt{c_\iota A_0}$, and at least one has modulus at least $\delta_s/\sqrt{m}$ at every point, because $|\mathbf{s}|^2 \leq m \max_j |\mathbf{s}_j|^2$. Theorem 4.5 gives a holomorphic map $\chi : \mathbb{D}_R \rightarrow (\mathbb{C}^m)^*$, whose components belong to the algebra on $\mathbb{D}_R$ obtained from $\mathfrak{A}$ by dilation, satisfying

$$\chi \cdot \mathbf{s} = 1, \quad \|\chi\|_\infty + \|\chi'\|_2 \leq C_\chi \quad (38)$$

for a constant $C_\chi = C_\chi(X, \omega, \iota, A_0, B_0)$. The dot denotes the bilinear pairing $\chi \cdot \mathbf{s} = \sum_{j=1}^m \chi_j \mathbf{s}_j$, without complex conjugation. The constant absorbs the fixed dimension factors in passing from scalar to vector norms.

To extend the lift past $\overline{\mathbb{D}}_R$, dilate χ radially while keeping $\mathbf{s}$ fixed, then restore their pairing by division. For $0 < \rho < 1$ sufficiently close to 1, set

$$\chi_\rho(z) := \chi(\rho z), \quad \gamma_\rho := \chi_\rho \cdot \mathbf{s}, \quad \hat{\chi} := \frac{\chi_\rho}{\gamma_\rho}.$$

Since $\chi(\rho z) \cdot \mathbf{s}(\rho z) = 1$, we have

$$\gamma_\rho(z) - 1 = \chi(\rho z) \cdot (\mathbf{s}(z) - \mathbf{s}(\rho z)).$$

The fixed lift $\mathbf{s}$ is uniformly continuous on $\overline{\mathbb{D}}_R$, and χ is bounded. So $\gamma_\rho \rightarrow 1$ uniformly there as $\rho \uparrow 1$, and we may require $|\gamma_\rho| \geq 1/2$. The quotient is therefore holomorphic near $\overline{\mathbb{D}}_R$ and satisfies $\hat{\chi} \cdot \mathbf{s} = 1$ and $\|\hat{\chi}\|_\infty \leq 2C_\chi$. Dilation gives $\|\chi'_\rho\|_2 \leq C_\chi$; the product and quotient rules give

$$\begin{aligned} \|\gamma'_\rho\|_2 &\leq C_\chi(1 + \sqrt{c_\iota A_0}), \\ \|\hat{\chi}'\|_2 &\leq 2C_\chi + 4C_\chi^2(1 + \sqrt{c_\iota A_0}) =: D_\chi. \end{aligned} \quad (39)$$

All norms here are on $\mathbb{D}_R$. The bound depends only on the fixed target, embedding, and A_0, B_0 ; the base lift $\mathbf{s}$ is unchanged.

4.2. Uniform lifts and holomorphic variations

The affine model below is a homogeneous version of the classical incidence-complement construction known as the Jouanolou trick; see [17, §2.6, p. 771]. The pairs $(\mathbf{s}, \chi)$ lie in this fixed model, whose defining equation $\chi \cdot \mathbf{s} = 1$ excludes the cone vertex, and a holomorphic right inverse lifts tangent fields into it. The model depends only on the embedding; the area and weighted-derivative caps determine a compact subset and a retraction near it.

Lemma 4.6. *Let X be a smooth projective manifold, and fix a projective embedding $\iota : X \hookrightarrow \mathbb{P}^{m-1}$. There are a closed Stein submanifold $\mathcal{Y} \subset \mathbb{C}^{2m}$, a holomorphic submersion $\pi_X : \mathcal{Y} \rightarrow X$, and a holomorphic bundle map $\mathcal{L} : \pi_X^* T_X \rightarrow T_{\mathcal{Y}}$ such that $d\pi_X \circ \mathcal{L} = \text{Id}$, all depending only on ι .*

Proof. Consider the affine cone of the embedding,

$$\widehat{X} := \{0\} \cup \{\mathbf{s} \in \mathbb{C}^m \setminus \{0\} : [\mathbf{s}] \in \iota(X)\},$$

and set

$$\mathcal{Y} := \{(\mathbf{s}, \chi) \in \widehat{X} \times (\mathbb{C}^m)^* : \chi \cdot \mathbf{s} = 1\}, \quad \pi_X(\mathbf{s}, \chi) := \iota^{-1}[\mathbf{s}]. \quad (40)$$

The defining equation excludes the cone vertex. On a chart $\mathbf{s}_j \neq 0$, local coordinates consist of coordinates on X , a nonzero scale factor, and $m - 1$ unconstrained dual coordinates. Hence $\mathcal{Y}$ is a smooth closed affine algebraic variety and therefore Stein, while π_X is a holomorphic submersion.

For the holomorphic splitting, set $V := \mathbb{C}^m$ and let $\ell := \iota^* \mathcal{O}_{\mathbb{P}^{m-1}}(-1)$ be the tautological line bundle on X ; its fiber $\ell_x \subset V$ is the line representing $\iota(x)$. Define

$$\mathcal{U} := \{(x, \chi) \in X \times V^* : \chi|_{\ell_x} \neq 0\}.$$

The map

$$\Theta : \mathcal{Y} \longrightarrow \mathcal{U}, \quad (\mathbf{s}, \chi) \longmapsto (\pi_X(\mathbf{s}, \chi), \chi),$$

is biholomorphic. Indeed, if s_0 is a local nonvanishing holomorphic section of ℓ , then

$$\sigma(x, \chi) := \frac{s_0(x)}{\chi(s_0(x))}$$

is independent of the choice of s_0 , is holomorphic, lies in ℓ_x , and satisfies $\chi(\sigma(x, \chi)) = 1$. Thus

$$\Theta^{-1}(x, \chi) = (\sigma(x, \chi), \chi).$$

Since $\mathcal{U}$ is open in $X \times V^*$, its tangent space at (x, χ) is $T_x X \oplus V^*$. Transporting the vectors $(v, 0)$ by $d\Theta^{-1}$ gives the required holomorphic right inverse

$$\mathcal{L} : \pi_X^* T_X \rightarrow T_{\mathcal{Y}}, \quad \mathcal{L}_y(v) := d(\Theta^{-1})_{\Theta(y)}(v, 0), \quad d\pi_X \circ \mathcal{L} = \text{Id}. \quad (41)$$

□

The holomorphic variations are built from three pieces of data: a controlled boundary-unitary frame, a lift of the disk, and lifts of the frame vectors. Under fixed area and weighted-derivative caps, the lifted disks lie in a common compact subset of the fixed affine model, and the estimates for all three ingredients are uniform.

Theorem 4.7. *Let (X, ω) be a compact projective manifold of complex dimension $n \geq 1$ with a smooth Hermitian metric, and fix a projective embedding $\iota : X \hookrightarrow \mathbb{P}^{m-1}$. Fix the affine model $\mathcal{Y}$, the submersion π_X , and the right inverse $\mathcal{L}$ supplied by Lemma 4.6. Given $A_0, B_0 \geq 0$, one can choose a compact set $K_{A_0, B_0} \Subset \mathcal{Y}$, an open neighborhood Ω of K_{A_0, B_0} in $\mathbb{C}^{2m}$, a holomorphic map $\tilde{\pi}_X : \Omega \rightarrow X$ with $\tilde{\pi}_X|_{\mathcal{Y} \cap \Omega} = \pi_X$, and constants*

$$\Gamma, D_1, D_{\tilde{f}}, C_{\theta, \infty}, D_{\theta} < \infty.$$

These cap-dependent choices depend only on $(X, \omega, \iota, A_0, B_0)$ and are independent of the source radius and the individual disk; the affine model and its splitting depend only on ι . For every $R > 0$ and every map f holomorphic on a neighborhood of $\overline{\mathbb{D}}_R$ with $\text{Area}_R(f) \leq A_0$ and $B_R(f) \leq B_0$, one can choose a frame e , a lifted disk $\tilde{f}$, and a matrix θ compatibly so that the following properties hold:

- (i) *e is a holomorphic frame of $f^* T_X$, smooth on the closed disk and unitary on its boundary, whose Gram matrix $G(z) := (\langle e_j(z), e_k(z) \rangle_{\omega})_{j,k=1}^n$ satisfies*

$$\Gamma^{-1} \text{Id}_n \leq G \leq \Gamma \text{Id}_n, \quad \sum_{a=1}^n \int_{\mathbb{D}_R} |\nabla_z e_a|^2 dA \leq D_1;$$

- (ii) *$\tilde{f}$ is a map from a neighborhood of $\overline{\mathbb{D}}_R$ to $\mathcal{Y}$, holomorphic on that neighborhood, such that*

$$\pi_X \circ \tilde{f} = f, \quad \tilde{f}(\overline{\mathbb{D}}_R) \subset K_{A_0, B_0}, \quad \int_{\mathbb{D}_R} |\tilde{f}'|^2 dA \leq D_{\tilde{f}};$$

(iii) the frame vectors have holomorphic lifts $\theta_1, \dots, \theta_n : \mathbb{D}_R \rightarrow \mathbb{C}^{2m}$ along $\tilde{f}$. Their column matrix and the lifting identities are

$$\begin{aligned} \theta &:= (\theta_1, \dots, \theta_n) : \mathbb{D}_R \rightarrow \text{Mat}_{2m \times n}(\mathbb{C}), \\ d(\tilde{\pi}_X)_{\tilde{f}(z)} \theta_a(z) &= e_a(z) \quad (1 \leq a \leq n). \end{aligned} \tag{42}$$

The matrix θ is smooth on the closed disk and satisfies

$$\sup_{z \in \mathbb{D}_R} \|\theta(z)\|_{\text{op}} \leq C_{\theta, \infty}, \quad \int_{\mathbb{D}_R} \|\theta'\|_{\text{HS}}^2 dA \leq D_{\theta}.$$

Proof. Fix a disk f satisfying the caps A_0, B_0 .

(i) Choose $R' > R$ so that $\overline{\mathbb{D}_{R'}}$ lies in the given source neighborhood, and choose a holomorphic frame $\tilde{e}$ of f^*T_X on $\mathbb{D}_{R'}$; such a frame exists by the Oka–Grauert principle [20, Satz 6, p. 270], since the source is contractible and Stein. Restrict its Gram matrix to $\partial\mathbb{D}_R$ and apply Lemma 4.1 to $H(\zeta) = \overline{G(R\zeta)}$. For the resulting factor A , set $e(z) := \tilde{e}(z)A(z/R)$. Conjugating the factorization identity gives its boundary Gram matrix $A(z/R)^T \tilde{G}(z) \overline{A(z/R)} = \text{Id}_n$. e is therefore holomorphic, smooth on the closed disk, and unitary on its boundary. Proposition 4.3 gives (34) and (35), proving (i).

(ii) By Lemma 4.4, f has a homogeneous lift $\mathbf{s}$ with

$$|\mathbf{s}| \leq 1, \quad |\mathbf{s}| \geq \delta_{\mathbf{s}}, \quad \int_{\mathbb{D}_R} |\mathbf{s}'|^2 dA \leq c_l A_0.$$

Its coordinates satisfy the hypotheses of Theorem 4.5 with algebra-norm bound $C_{\mathbf{s}}$ and corona lower bound $\delta_{\mathbf{s}}/\sqrt{m}$. Thus (38) supplies a vector χ satisfying $\chi \cdot \mathbf{s} = 1$, and the dilation construction leading to (39) gives a vector $\hat{\chi}$, holomorphic near the closed disk, with

$$\hat{\chi} \cdot \mathbf{s} = 1, \quad \|\hat{\chi}\|_{L^\infty(\mathbb{D}_R)} \leq 2C_{\chi}, \quad \|\hat{\chi}'\|_{L^2(\mathbb{D}_R)} \leq D_{\chi}.$$

The identity $\hat{\chi} \cdot \mathbf{s} = 1$ shows that $\tilde{f} := (\mathbf{s}, \hat{\chi})$ takes values in the fixed space $\mathcal{Y}$ of (40). It satisfies

$$\tilde{f}(\overline{\mathbb{D}_R}) \subset K_{A_0, B_0} := \mathcal{Y} \cap \{|\mathbf{s}| \leq 1, |\chi| \leq 2C_{\chi}\}, \quad \int_{\mathbb{D}_R} |\tilde{f}'|^2 dA \leq c_l A_0 + D_{\chi}^2 =: D_{\tilde{f}}. \tag{43}$$

The set K_{A_0, B_0} is compact: it is closed and bounded, and $\chi \cdot \mathbf{s} = 1$ prevents $\mathbf{s}$ from approaching the cone vertex.

Since $\mathcal{Y}$ is a closed Stein submanifold of $\mathbb{C}^{2m}$, the Docquier–Grauert tubular neighborhood theorem [11, Satz 3], in the holomorphic deformation-retraction formulation [16, Theorem 3.1], gives a holomorphic retraction $\mathbf{r} : \Omega \rightarrow \mathcal{Y} \cap \Omega$ from a neighborhood Ω of K_{A_0, B_0} , equal to the identity on $\mathcal{Y} \cap \Omega$. Its derivatives are bounded on smaller compact neighborhoods of K_{A_0, B_0} . Set $\tilde{\pi}_X := \pi_X \circ \mathbf{r}$. Since $\pi_X \circ \tilde{f} = f$ and $\tilde{\pi}_X|_{\mathcal{Y} \cap \Omega} = \pi_X$, (43) proves (ii). The choices of K_{A_0, B_0} , Ω , and $\tilde{\pi}_X$ depend only on the fixed target data and the two caps; they are fixed before any individual lifted disk $\tilde{f}$ is chosen. The space $\mathcal{Y}$, the projection π_X , and the splitting $\mathcal{L}$ were already fixed from the embedding alone.

(iii) Regard the splitting $\mathcal{L}$ from (41) as a bundle map from $\pi_X^*T_X$ to the ambient trivial bundle $\mathcal{Y} \times \mathbb{C}^{2m}$. In $\nabla\mathcal{L}$, use the pullback Chern connection on $\pi_X^*T_X$ and the flat connection on this ambient bundle. Let

$$L_0 := \sup_{K_{A_0, B_0}} \|\mathcal{L}\|, \quad L_1 := \sup_{K_{A_0, B_0}} \|\nabla\mathcal{L}\|.$$

Set

$$\theta_a(z) := \mathcal{L}_{\tilde{f}(z)} e_a(z).$$

Then $d(\tilde{\pi}_X)_{\tilde{f}} \theta_a = e_a$. With the preceding connection convention, ambient differentiation gives

$$\theta'_a = \mathcal{L}_{\tilde{f}} \nabla_z e_a + (\nabla_{\tilde{f}'} \mathcal{L}) e_a. \tag{44}$$

The frame and lifted-disk bounds, with $|a + b|^2 \leq 2|a|^2 + 2|b|^2$, give

$$\|\theta\|_{L^\infty, \text{op}} \leq L_0 \sqrt{\Gamma} =: C_{\theta, \infty}, \quad (45)$$

$$\int_{\mathbb{D}_R} \|\theta'\|_{\text{HS}}^2 dA \leq 2L_0^2 D_1 + 2n\Gamma L_1^2 D_{\tilde{f}} =: D_\theta. \quad (46)$$

This proves (iii). $\square$

We now add the lifted variation fields and project to X . The lifted frame matrix θ is holomorphic on $\mathbb{D}_R$ but need not extend holomorphically across $\partial\mathbb{D}_R$. For $0 < \rho < 1$, we therefore replace $\theta(z)$ in the variation field by $\theta(\rho z)$. The new matrix is holomorphic on the larger disk $\mathbb{D}_{R/\rho}$, so the resulting field extends past $\overline{\mathbb{D}}_R$. We leave the central lift $\tilde{f}(z)$ unchanged; hence the projected family at parameter $t = 0$ is still the original map f . The multiplier is also unchanged, so its vanishing orders at the origin are preserved. As $\rho \uparrow 1$, the regularized fields converge to the original fields, and their paired trace converges to the value computed in Section 3. The parameter range and cubic remainder remain uniform throughout this approximation.

The embedding fixes $\mathcal{Y}, \pi_X, \mathcal{L}$, independently of the disks and caps, and the caps A_0, B_0 then fix $K_{A_0, B_0}, \Omega, \tilde{\pi}_X$; these data, together with D_Φ, N_Φ , determine t_0, C_T . Each disk comes with its own compatible frame, lift, and lifted fields, as in Theorem 4.7. The approximation parameter ρ may depend on this triple, on Φ , and on the prescribed trace error ε_{tr} , whereas t_0 and C_T are independent of R, f, Φ, ρ , and ε_{tr} ; the extension neighborhood may depend on the family, on Φ , and on ρ , and its width need not be uniform.

Theorem 4.8. *Let X be a compact projective manifold of dimension $n \geq 1$, equipped with a Kähler metric ω . Fix the projective embedding $\iota : X \hookrightarrow \mathbb{P}^{m-1}$, and the bounds $A_0, B_0 \geq 0$ as in Theorem 4.7. Let $D_\Phi \geq 0$, and let $N_\Phi \geq 1$ be an integer. There are constants $t_0 > 0$ and $C_T < \infty$, depending only on $(X, \omega, \iota, A_0, B_0, D_\Phi, N_\Phi)$, with the following property.*

Let $R > 0$, and let $f : \mathbb{D}_R \rightarrow X$ extend holomorphically to a neighborhood of $\overline{\mathbb{D}}_R$ and satisfy $\text{Area}_R(f) \leq A_0$ and $B_R(f) \leq B_0$. Use the fixed K_{A_0, B_0}, Ω , and $\tilde{\pi}_X$ from Theorem 4.7, and choose any compatible disk-dependent triple $e, \tilde{f}, \theta$ supplied by that theorem. Suppose that $\Phi = (\Phi_1, \dots, \Phi_{N_\Phi}) : \mathbb{D}_R \rightarrow \text{Mat}_{n \times N_\Phi}(\mathbb{C})$ extends holomorphically to an open neighborhood of $\overline{\mathbb{D}}_R$ and satisfies

$$\sup_{z \in \overline{\mathbb{D}}_R} \|\Phi(z)\|_{\text{op}} \leq 1, \quad \int_{\mathbb{D}_R} \|\Phi'\|_{\text{HS}}^2 dA \leq D_\Phi,$$

*where the column Φ_a determines the section $e\Phi_a = \sum_{j=1}^n e_j \Phi_{ja}$ of f^*T_X . The constants t_0 and C_T are independent of R, f, Φ and of this compatible triple.*

For each $0 < \rho < 1$, write

$$\Xi_\rho(z) := \theta(\rho z)\Phi(z), \quad (\Xi_\rho)_a(z) := \sum_{j=1}^n \theta_j(\rho z)\Phi_{ja}(z).$$

(i) *Holomorphic realization. For every $0 < \rho < 1$, each column defines the family*

$$f_{a,t}^\rho(z) := \tilde{\pi}_X(\tilde{f}(z) + t(\Xi_\rho)_a(z)), \quad t \in \mathbb{C}, \quad |t| \leq t_0, \quad 1 \leq a \leq N_\Phi.$$

All these families are jointly holomorphic on one open neighborhood of $\overline{\mathbb{D}}_R \times \{t \in \mathbb{C} : |t| \leq t_0\}$; for every $0 < \rho < 1$ that neighborhood may depend on the selected disk-dependent triple, Φ , and ρ , but not on the column a .

(ii) *Uniform cubic remainder. For every $0 < \rho < 1$, $1 \leq a \leq N_\Phi$, and $\tau \in \{1, i\}$, write $t = \tau s$ with $s \in \mathbb{R}$ and set*

$$\ell_{a,\tau}^\rho := \left. \frac{d}{ds} \right|_{s=0} \text{Area}_R(f_{a,\tau s}^\rho), \quad H_{a,\tau}^\rho := \left. \frac{d^2}{ds^2} \right|_{s=0} \text{Area}_R(f_{a,\tau s}^\rho).$$

For $|s| \leq t_0$,

$$\text{Area}_R(f_{a,\tau s}^\rho) \leq \text{Area}_R(f) + \ell_{a,\tau}^\rho s + \frac{1}{2} H_{a,\tau}^\rho s^2 + C_T |s|^3 \quad (47)$$

with the same constant C_T for both choices of τ .

- (iii) Trace approximation. If Φ is boundary coisometric, meaning $\Phi\Phi^* = \text{Id}_n$ on $\partial\mathbb{D}_R$, then for every $\varepsilon_{\text{tr}} > 0$ there is a number

$$\rho_{\text{tr}} = \rho_{\text{tr}}(R, f, e, \tilde{f}, \theta, \Phi, \varepsilon_{\text{tr}}) \in (0, 1)$$

such that

$$\left| \sum_{a=1}^{N_\Phi} (H_{a,1}^{\rho_{\text{tr}}} + H_{a,i}^{\rho_{\text{tr}}}) - \left(4 \int_{\mathbb{D}_R} \|\Phi'\|_{\text{HS}}^2 dA - 2 \int_{\mathbb{D}_R} f^* \text{Ric}(\omega) \right) \right| \leq \varepsilon_{\text{tr}}. \quad (48)$$

Only the trace approximation requires choosing ρ_{tr} close to 1; the constants t_0 and C_T remain independent of ε_{tr} and ρ_{tr} .

Proof. Use the fixed cap-dependent ambient data and the selected compatible triple from Theorem 4.7. Set

$$\delta_0 := \min\{1, \text{dist}(K_{A_0, B_0}, \mathbb{C}^{2m} \setminus \Omega)\} > 0,$$

and choose a compact $K' \Subset \Omega$ whose interior contains the closed Euclidean $3\delta_0/4$ -thickening of K_{A_0, B_0} . In the standard coordinate frame of $\mathbb{C}^{2m}$, let g_0 be the positive-semidefinite Hermitian coefficient matrix of $\tilde{\pi}_X^* \omega$ on K' , so

$$v^\top g_0(w) \bar{v} = |d(\tilde{\pi}_X)_w(v)|_\omega^2$$

for a coefficient column v . Only smooth coefficient bounds are used; the pullback form may be degenerate along the fibers. Take $C_{g,3}$ to bound the coefficients and their derivatives through order three on K' .

Take a multiplier Φ satisfying the stated bounds and, for $0 < \rho < 1$, set $\Xi_\rho := \theta(\rho \cdot) \Phi$. Dilation gives $\int_{\mathbb{D}_R} |\rho \theta'(\rho z)|^2 dA(z) = \int_{\mathbb{D}_{\rho R}} |\theta'(w)|^2 dA(w) \leq D_\theta$. Use this identity, $\|\Phi\|_\infty \leq 1$, and $(\theta(\rho z) \Phi)' = \rho \theta'(\rho z) \Phi + \theta(\rho z) \Phi'$ to obtain

$$\sup_{z \in \mathbb{D}_R} \|\Xi_\rho(z)\|_{\text{op}} \leq C_{\theta, \infty}, \quad \int_{\mathbb{D}_R} \|\Xi'_\rho\|_{\text{HS}}^2 dA \leq 2D_\theta + 2C_{\theta, \infty}^2 D_\Phi =: D_\Xi. \quad (49)$$

Choose

$$t_0 := \min \left\{ 1, \frac{\delta_0}{4 \max\{1, C_{\theta, \infty}\}} \right\}.$$

For $|t| \leq 2t_0$, the displacement of each column perturbation is at most $2t_0 C_{\theta, \infty} \leq \delta_0/2$. So $\tilde{f} + t(\Xi_\rho)_a$ stays in K' , uniformly in the disk and the column.

(i) Fix a column and write $\Xi := (\Xi_\rho)_a$. The function $\theta(\rho z)$ is holomorphic for $|z| < R/\rho$; since $\tilde{f}$ and Φ extend past the closed disk, so does $\tilde{f} + t\Xi$. On the closed disk and for $|t| \leq t_0$ its image lies in the closed $\delta_0/2$ -thickening of K_{A_0, B_0} , which is compactly contained in $\text{int } K' \Subset \Omega$. Compactness gives a neighborhood of $\overline{\mathbb{D}_R} \times \{|t| \leq t_0\}$ on which the image remains in Ω , so the projected family is jointly holomorphic there. Intersect these neighborhoods over the finitely many columns.

(ii) In ambient coordinates write $w(t) := \tilde{f} + t\Xi$ and $v(t) := \tilde{f}' + t\Xi'$. The area density is $v(t)^\top g_0(w(t)) \overline{v(t)}$. Along either real parameter axis $t = \tau s$, with $\tau \in \{1, i\}$, the functions $w(\tau s)$ and $v(\tau s)$ are affine in s , with first derivatives $\tau\Xi$ and $\tau\Xi'$. For fixed z , put $G(s) := g_0(w(\tau s))$, $V(s) := v(\tau s)$, and $Q := \tau\Xi'$. Ordinary differentiation in the real variable s gives

$$\frac{d^3}{ds^3} (V^\top G \bar{V}) = V^\top G''' \bar{V} + 3Q^\top G'' \bar{V} + 3V^\top G'' \bar{Q} + 6Q^\top G' \bar{Q}.$$

The chain rule bounds $|G^{(k)}(s)|$ by $C_{g,3}|\Xi|^k$ for $1 \leq k \leq 3$, up to fixed dimension factors. The bounds $\|\Xi\|_\infty \leq C_{\theta,\infty}$ and $|t| \leq t_0$ consequently give the integrable estimate

$$\left| \frac{d^3}{ds^3} (v(\tau s)^\top g_0(w(\tau s)) \overline{v(\tau s)}) \right| \leq C(C_{g,3}, C_{\theta,\infty}, t_0) (|\tilde{f}'|^2 + |\Xi'|^2).$$

The right-hand side has a uniformly bounded integral by (43) and (49). The third derivative of the area is therefore uniformly bounded along both real parameter axes. Taylor's theorem gives (47) with one remainder constant C_T .

(iii) The tangent fields of the actual families satisfy, for each fixed f ,

$$\xi_{\rho,a} := d(\tilde{\pi}_X)_{\tilde{f}}(\theta(\rho \cdot) \Phi_a) \longrightarrow e\Phi_a \quad \text{in } C^1(\overline{\mathbb{D}}_R) \quad (\rho \uparrow 1),$$

because $\theta(\rho \cdot) \rightarrow \theta$ in the same norm. The Kähler condition enters at this point, through Lemma 3.3. If Φ is boundary coisometric, that lemma, continuity of (24), and Proposition 3.4 therefore give

$$\sum_a \mathcal{H}_{R,f}(\xi_{\rho,a}) \longrightarrow 4 \int_{\mathbb{D}_R} \|\Phi'\|_{\text{HS}}^2 dA - 2 \int_{\mathbb{D}_R} f^* \text{Ric}(\omega).$$

The summand is $H_{a,1}^\rho + H_{a,i}^\rho$ by Lemma 3.3. Choose ρ_{tr} close enough to one to obtain the prescribed error. Parts (i)–(ii) and (49) hold for every $\rho \in (0, 1)$ with the same t_0, C_T . Their dependence is only on the fixed target data and A_0, B_0, D_Φ, N_Φ . $\square$

4.3. A common area and derivative estimate

For the fixed caps, write $K = K_{A_0, B_0}$, $\Pi = \tilde{\pi}_X$, and $D = D_{\tilde{f}}$ for the common compact set, ambient projection, and Dirichlet bound supplied by Theorem 4.7. Each compatible lift H satisfies

$$\Pi H = f, \quad H(\overline{\mathbb{D}}_R) \subset K, \quad \int_{\mathbb{D}_R} |H'|^2 dA \leq D. \quad (50)$$

Choose $d_K > 0$ such that the closed Euclidean d_K -neighborhood K_{d_K} of K is compactly contained in Ω . Write $g(w)$ for the Hermitian coefficient matrix of $\Pi^* \omega$ in ambient coordinates. Thus

$$a^\top g(w) \bar{a} = |d\Pi_w(a)|_\omega^2.$$

The form can be degenerate. On K_{d_K} , fix bounds

$$\|g(w)\|_{\text{op}} \leq M_0, \quad \|Dg(w)[v]\|_{\text{op}} \leq L_g |v|, \quad (51)$$

where Dg is the real derivative in ambient coordinates. These bounds are fixed before selecting an individual disk.

Lemma 4.9. *Let $R > 0$, $N \geq 1$, and $E \geq 0$. If $F : \mathbb{D}_R \rightarrow \mathbb{C}^N$ is holomorphic and $\int_{\mathbb{D}_R} |F'|^2 dA \leq E$, then*

$$(R - |z|)|F'(z)| \leq \sqrt{E/\pi} \quad (z \in \mathbb{D}_R).$$

Proof. For $0 < r < R - |z|$, the classical submean inequality [9, Ch. I, §4], applied to $|F'|^2$, gives

$$\pi r^2 |F'(z)|^2 \leq \int_{\mathbb{D}_r(z)} |F'|^2 dA \leq E.$$

Let $r \uparrow R - |z|$. $\square$

Proposition 4.10. *Let (X, ω) be a compact projective manifold with a smooth Hermitian metric. Fix $R > 0$ and the cap-dependent ambient data $K, \Omega, \Pi, D, d_K, M_0, L_g$ from (50)–(51). Let H be holomorphic near $\overline{\mathbb{D}}_R$ and satisfy (50), with $f = \Pi \circ H$. Let $S, E \geq 0$, and let Ξ be a $\mathbb{C}^{2m}$ -valued map holomorphic near $\mathbb{D}_R$, with*

$$\|\Xi\|_\infty \leq S, \quad \int_{\mathbb{D}_R} |\Xi'|^2 dA \leq E.$$

Fix $0 < \bar{t} \leq 1$ with $\bar{t}S < d_K$. For $|t| \leq \bar{t}$, set $f_t = \Pi(H + t\Xi)$ and

$$C_A = L_gSD + 2M_0\sqrt{DE} + M_0\bar{t}E. \quad (52)$$

Then

$$|\text{Area}_R(f_t) - \text{Area}_R(f)| \leq C_A|t|, \quad (53)$$

$$|B_R(f_t)^2 - B_R(f)^2| \leq \frac{C_A}{\pi}|t|. \quad (54)$$

The same family has the pointwise displacement bound

$$\text{dist}_\omega(f_t(z), f(z)) \leq \sqrt{M_0}S|t| \quad (z \in \mathbb{D}_R). \quad (55)$$

The estimate for B_R^2 gives both an upper and a lower bound for the new value of B_R :

$$\sqrt{\max\{0, B_R(f)^2 - C_A|t|/\pi\}} \leq B_R(f_t) \leq \sqrt{B_R(f)^2 + C_A|t|/\pi}. \quad (56)$$

Proof. The entire segment $H(z) + st\Xi(z)$, $0 \leq s \leq 1$, lies in K_{d_K} , so (51) applies along it. Put $a = H'(z)$, $b = \Xi'(z)$, and $g_t = g(H(z) + t\Xi(z))$. The change in area density has three contributions: the change in the metric coefficients, a mixed derivative term, and a quadratic derivative term. More precisely,

$$(a + tb)^\top g_t \bar{a} + t\bar{b} - a^\top g_0 \bar{a} = a^\top (g_t - g_0) \bar{a} + 2\text{Re}(tb^\top g_t \bar{a}) + |t|^2 b^\top g_t \bar{b}. \quad (57)$$

Its absolute value is at most

$$|t|L_gS|a|^2 + 2M_0|t||a||b| + M_0|t|^2|b|^2.$$

Integrating, and applying Cauchy–Schwarz to the middle term, proves (53).

For the second estimate, put $d = R - |z|$. Lemma 4.9 gives $d|a| \leq \sqrt{D/\pi}$ and $d|b| \leq \sqrt{E/\pi}$. Multiplying the same pointwise estimate by d^2 therefore bounds the difference of the two squared weighted derivatives by $C_A|t|/\pi$. Use $|\sup p - \sup q| \leq \sup |p - q|$ and $B_R(f)^2 = \sup d^2|f_z|_\omega^2$ to obtain (54). Finally, the path $s \mapsto \Pi(H(z) + st\Xi(z))$, $0 \leq s \leq 1$, has speed at most $\sqrt{M_0}|t||\Xi(z)|$. Its length proves (55). Only bounds on the possibly degenerate metric coefficients have been used. $\square$

4.4. Uniform descent and enlargement

Return now to the relative setting, where $\text{Ric}(\omega) \geq \kappa\omega$. The trace is negative whenever $\text{Area}_R(f) > A_*$. Selecting one real direction and the sign of its parameter gives descent; the common parameter radius and Taylor remainder give a uniform step size under fixed area and weighted-derivative bounds.

Proof of Proposition 2.1. Put $x_0 = f(0)$ and $\lambda = \lambda_{x_0}^F$.

For the multiplier (27), one has $N_\Phi = n$, $D_\Phi = \pi(n+1)$, and, by (29),

$$4\pi(n+1) - 2\kappa \text{Area}_R(f) \leq -2\kappa\delta. \quad (58)$$

Apply Theorem 4.8 with trace error $\varepsilon_{\text{tr}} = \kappa\delta$, choose the value ρ_{tr} supplied by part (iii), and abbreviate $f_{a,t} = f_{a,t}^{\rho_{\text{tr}}}$, $\ell_{a,\tau} = \ell_{a,\tau}^{\rho_{\text{tr}}}$, and $H_{a,\tau} = H_{a,\tau}^{\rho_{\text{tr}}}$. For every column a and every $|t| \leq t_0$, the identities $\Phi_a(0) = 0$ and $d(\tilde{\pi}_X)_{\tilde{f}(0)}\theta(0) = e(0)$ give the exact interpolation formula

$$\begin{aligned} f_{a,t}(0) &= f(0), \\ (f_{a,t})_z(0) &= f_z(0) + t e(0) \Phi'_a(0), \\ \lambda((f_{a,t})_z(0)) &= \lambda(f_z(0)) + t \lambda(e(0) \Phi'_a(0)) = 1. \end{aligned} \quad (59)$$

Indeed, the ambient perturbation at $z = 0$ equals $\tilde{f}(0)$ for every t , so the chain rule always evaluates $d\tilde{\pi}_X$ at the same point $\tilde{f}(0)$. Moreover, the product rule gives $(\Xi_{\rho_{\text{tr}}})'_a(0) = \theta(0)\Phi'_a(0)$, so radial regularization does not alter the derivative of the increment at the origin. Formula (59) fixes the displayed scalar component, not the entire first jet. For this choice, part (iii) and

(58) give $\sum_a (H_{a,1} + H_{a,i}) \leq -\kappa\delta$. One of the $2n$ real axes therefore has second area derivative $H_{a,\tau} \leq -\kappa\delta/(2n)$. Choose the sign of s so that $\ell_{a,\tau}s \leq 0$. Then (47) gives

$$\text{Area}_R(f_{a,\tau s}) - \text{Area}_R(f) \leq -\frac{\kappa\delta}{4n}s^2 + C_T|s|^3.$$

Before selecting the step, decrease the engine parameter radius to $\min\{t_0, 1, d_K/(2\max\{1, C_{\theta,\infty}\})\}$ if necessary, so that the selected perturbation segment lies in the compact tube used in Proposition 4.10. Set

$$t_* := \min\left\{\frac{t_0}{2}, \frac{\kappa\delta}{8n\max\{C_T, 1\}}\right\}, \quad \sigma := \frac{\kappa\delta}{8n}t_*^2. \quad (60)$$

Take $s \in \{t_*, -t_*\}$ with the chosen sign and set $g = f_{a,\tau s}$. The Taylor estimate gives $\text{Area}_R(g) \leq \text{Area}_R(f) - \sigma$. The selected increment has the bounds in (49). Proposition 4.10 gives the remaining estimate in (7) with

$$\beta_I = C_{A,I}t_*/\pi,$$

where $C_{A,I}$ is (52) for this increment. The parameter choice keeps its entire segment in the common tube. The constants $t_0, C_T, C_{A,I}$ depend only on the fixed caps and target data. The multiplier has the same cost for every nonzero covector $\lambda_{x_0}^F$. Thus t_*, σ, β_I are independent of the radius, the disk, and the center in F . $\square$

For scalar enlargement, it suffices to lift one finite-area disk and perturb its lift in one constant ambient direction.

Proof of Proposition 2.2. (i) Suppose that h extends holomorphically past $\mathbb{D}_R$. Theorem 4.7(ii) supplies a lift H defined near $\mathbb{D}_R$, with $H(\mathbb{D}_R) \subset K_{A_0, B_0}$, $\Pi \circ H = h$, and $\int_{\mathbb{D}_R} |H'|^2 dA \leq D_{\tilde{f}}$; here $\Pi = \tilde{\pi}_X$ and the compact set, ambient neighborhood, and Dirichlet bound depend only on the caps and fixed target data. Enlarge $D = D_{\tilde{f}}$ to at least 1 and put

$$Q = \sqrt{D/\pi}, \quad \bar{t} = \min\{1, d_K/(2\max\{1, Q\})\},$$

where d_K is the common tube radius used in Section 4.3. For $v = H'(0)$, Lemma 4.9 gives

$$R|v| \leq Q, \quad \|zv\|_\infty \leq Q, \quad \int_{\mathbb{D}_R} |(zv)'|^2 dA = \pi R^2 |v|^2 \leq D.$$

The affine formula (8) therefore stays in the common tube and extends past the closed disk. Proposition 4.10 gives the area and B^2 bounds with

$$C_{II} = D(L_g Q + 3M_0). \quad (61)$$

Source dilation satisfies the exact identities

$$\text{Area}_{(1+t)R}(G_t) = \text{Area}_R(h_t), \quad B_{(1+t)R}(G_t) = B_R(h_t).$$

The affine increment vanishes at 0 and gives $(h_t)_z(0) = (1+t)h_z(0)$; dilation restores the entire derivative. The displacement estimate refers to corresponding source points:

$$\text{dist}_\omega(G_t((1+t)z), h(z)) \leq \sqrt{M_0} Q t.$$

(ii) Now let h be given only on $\mathbb{D}_R$. For a chosen t , the restriction radius $r = (1+t/2)R/(1+t)$ is strictly smaller than R . Hence $h|_{\mathbb{D}_r}$ extends past its closed disk, and restriction gives

$$\text{Area}_r(h) \leq \text{Area}_R(h) \leq A_0, \quad B_r(h) \leq B_R(h) \leq B_0.$$

Apply the closed-disk construction at radius r ; its dilated map is exactly g in (11) and extends past $\mathbb{D}_{R'}$. Its two-sided estimates read

$$|\text{Area}_{R'}(g) - \text{Area}_r(h)| \leq C_{II}t, \quad |B_{R'}(g)^2 - B_r(h)^2| \leq C_{II}t/\pi.$$

The restriction inequalities give (12), while the closed-disk interpolation gives $g(0) = h(0)$ and $g_z(0) = h_z(0)$. The finite displacement bound compares $g((1+t)z)$ with $h(z)$ for $z \in \mathbb{D}_r$; it does not give a closeness estimate for the principal-map recovery. $\square$

5. COMPACTNESS, MARKED VALUES, AND FINITE-AREA DISKS

Here we prove the compactness and removal statements used above. Marked compactness is applied on one fixed interior disk, with all component areas counted together; a local small-area estimate supplies the weighted derivative bound for an individual finite-area disk, and point selection produces the sphere used in recovery.

Throughout this section, X is a compact complex manifold equipped with a smooth Hermitian metric ω .

5.1. Finite-area removal

Finite area fills the punctures of a principal limit, including infinity when its source is $\mathbb{C}$. We first prove removal so that the compactness statement can include the extended principal map.

With $g_\omega(v, w) = \omega(v, Jw)$, the energy convention in the compactness and removal results used below is

$$\|df\|_{L^2}^2 = \int (|f_x|_{g_\omega}^2 + |f_y|_{g_\omega}^2) dA = 2 \int f^* \omega = 2 \text{Area}(f), \quad (62)$$

because $f_y = Jf_x$. Thus their energy hypotheses are equivalent to our area hypotheses, with the fixed factor 2.

Proof of Theorem 2.4. Finite area gives $\int_{0 < |z| < r} f^* \omega \rightarrow 0$ as $r \downarrow 0$, so every sufficiently small concentric annulus has small energy by (62). Compactness of X supplies completeness of the target metric and uniform continuity of its complex structure. Corollary 1.6 of Ivashkovich and Shevchishin [28] therefore gives a continuous extension at the puncture. In a target chart about the limiting value, the coordinate functions extend holomorphically by the Riemann removable singularity theorem. For an entire map, $\zeta = 1/z$ identifies infinity with a puncture and preserves area, so the same argument applies there. See also [32, Theorem 4.2.1]. $\square$

For maps into $\mathbb{P}^1$, removal also follows from the great Picard theorem [39, §15, pp. 164–166]: an essential singularity would give infinitely many preimages of all but at most two values, and the area formula with multiplicities would force infinite spherical area.

5.2. Principal compactness

Spheres obtained by rescaling near energy concentration points go back to Sacks and Uhlenbeck [40, pp. 2–3 and Section 4]; Gromov’s compactness theory records how such components attach [21]. We use the stable convergence of Definition 2.5 and the compactness theorem with boundary in Theorem 1 of Ivashkovich and Shevchishin [28].

The *principal map* is the limit in the original source coordinate and may be constant. A *bubble* is a sphere obtained by rescaling; the finite-area extension of an entire principal map is also a sphere, but need not be a bubble.

Proof of Proposition 2.3. First pass to a subsequence whose total areas converge to the original lower limit A_∞ . A diagonal weak selection of the area measures on compact subdisks gives a Radon measure μ on $\mathbb{D}_{R_\infty}$. For every continuous compactly supported χ with $0 \leq \chi \leq 1$,

$$\int \chi d\mu = \lim_j \int \chi f_j^* \omega \leq A_\infty,$$

so $\mu(\mathbb{D}_{R_\infty}) \leq A_\infty$ by exhaustion. Fix $p > 2$ and an area threshold $\epsilon_0 > 0$ small enough to apply Lemma 1.1 and Corollary 1.3 of Ivashkovich and Shevchishin [28], using (62), and set

$$\Sigma_\infty = \{z : \mu(\{z\}) \geq \epsilon_0\}.$$

The mass bound makes this set finite. For $z \notin \Sigma_\infty$, choose a source disk with μ -null boundary and μ -mass less than ϵ_0 . Weak convergence puts its area below that threshold for all large j . The cited result gives strong local $W^{1,p}$ convergence on a smaller disk; Sobolev embedding gives uniform convergence, and target charts and Cauchy estimates give smooth convergence. A diagonal choice on a countable cover gives f_∞ on the complement of Σ_∞ in the original coordinate. Its local definitions agree because they are limits of the same subsequence. Smooth convergence and exhaustion give $\text{Area}(f_\infty) \leq A_\infty$. Theorem 2.4 fills the finitely many punctures without changing area. $\square$

5.3. Two marked values and incidence

Fixing the complete first jet at 0 does not ensure that a nonconstant bubble retains the prescribed tangent direction. In the following example, the marked value lies on the bubble, while that direction remains on the principal component. On the unit disk, with the product Fubini–Study metric, consider

$$\widehat{f}_j(z) := ([jz^2 : 1], [z : 1]) : \mathbb{D} \longrightarrow \mathbb{P}^1 \times \mathbb{P}^1.$$

The areas are uniformly bounded, and in the affine coordinates $([w : 1], [v : 1])$ the complete first jet is fixed:

$$\widehat{f}_j(0) = ([0 : 1], [0 : 1]), \quad (\widehat{f}_j)_z(0) = (0, 1).$$

Away from zero, the principal limit is

$$\widehat{f}_\infty(z) = ([1 : 0], [z : 1]).$$

Its extension has value $([1 : 0], [0 : 1])$ at zero, different from the marked value. Rescaling by $z = \zeta/\sqrt{j}$ instead gives

$$\widehat{f}_j(\zeta/\sqrt{j}) = ([\zeta^2 : 1], [\zeta/\sqrt{j} : 1]) \longrightarrow ([\zeta^2 : 1], [0 : 1]).$$

The prescribed second-factor direction remains on the residual principal component. Correspondingly, $B_1(\widehat{f}_j)/|(\widehat{f}_j)_z(0)|_\omega \rightarrow \infty$.

Marked and nodal values retain incidence. Lemma 2.5 gives a chain joining two compact target sets with controlled total area. It supplies the compactness alternative in Section 2 and the final passage to the explicit area bound.

Proof of Lemma 2.5. First select a subsequence with $\text{Area}_{R_j}(f_j) \rightarrow A_\infty = \liminf_j \text{Area}_{R_j}(f_j)$. Proposition 2.3 gives an entire principal map f_∞ and a finite concentration set Σ_∞ on a further subsequence.

Choose $1 < r < \rho < r'$ with $\Sigma_\infty \subset \mathbb{D}_r$. To retain the source coordinate in the limit, apply compactness to the graphs

$$\widehat{f}_j(z) = (z, f_j(z)) : \mathbb{D}_\rho \longrightarrow \mathbb{P}^1 \times X.$$

The product Hermitian metric gives a uniform area bound: the first factor adds only the fixed Fubini–Study area of $\mathbb{D}_\rho$. The maps are defined on $\mathbb{D}_{r'}$ for large j and converge smoothly on a collar of $\partial\mathbb{D}_\rho$. Since the source complex structures are fixed, Theorem 1 of Ivashkovich and Shevchishin [28] applies, and its boundary-collar conclusion allows the parameterizations to agree with the original ones on a smaller collar.

Let $\sigma_j : \Sigma \rightarrow \overline{\mathbb{D}}_\rho$ and $\sigma_\infty : \Sigma \rightarrow C_\infty$ be the parameterizations in Definition 2.5, and write $(p, v) : C_\infty \rightarrow \mathbb{P}^1 \times X$ for the limiting nodal map. For each $a \in \{0, 1\}$, choose $\xi_{j,a} \in \Sigma$ such that $\sigma_j(\xi_{j,a}) = a$. After passing to a further subsequence, $\xi_{j,a} \rightarrow \xi_a$ for $a = 0, 1$; set $q_a = \sigma_\infty(\xi_a)$. The uniform convergence in Definition 2.5 gives

$$(p, v)(q_a) = \lim_{j \rightarrow \infty} (\widehat{f}_j \circ \sigma_j)(\xi_{j,a}) = \lim_{j \rightarrow \infty} (a, f_j(a)) \in \{a\} \times K_a.$$

Thus $p(q_a) = a$ and $v(q_a) \in K_a$ for $a = 0, 1$.

The genus-zero domain has one bordered root C_0 and a finite tree of spheres. Since p takes values in $\overline{\mathbb{D}}_\rho \subset \mathbb{C}$, it is constant on every sphere. The unchanged outer collar gives $p(z) = z$ there. The

argument principle on C_0 consequently gives exactly one preimage, counted with multiplicity, for every point of $\mathbb{D}_\rho$. Thus $p : C_0 \rightarrow \mathbb{D}_\rho$ is biholomorphic. In this coordinate, convergence off Σ_∞ identifies $v|_{C_0}$ with $f_\infty|_{\mathbb{D}_\rho}$; holomorphic uniqueness extends the identification across the omitted points.

Choose compact regions in the interior of the root and in each nonconstant sphere, avoiding the nodes. A single stable parametrization gives orientation-preserving embeddings of these regions into $\mathbb{D}_\rho$ with pairwise disjoint images. Uniform and strong $W^{1,2}$ convergence give convergence of their ω -integrals. For any fixed $L > \rho$, these regions are also disjoint from the annulus $\mathbb{D}_L \setminus \overline{\mathbb{D}_\rho}$, where the original maps converge smoothly to f_∞ , since $\Sigma_\infty \subset \mathbb{D}_r$. All these regions lie in $\mathbb{D}_{R_j}$ for large j , so their limiting areas sum to at most A_∞ . Exhaust the root interior and the spheres away from the nodes, then let $L \rightarrow \infty$. This gives

$$\text{Area}(f_\infty) + \sum_{\alpha} \text{Area}(u_\alpha) \leq A_\infty, \quad (63)$$

where u_α are the nonconstant sphere maps. Only the original form ω is integrated, with mapping multiplicities; the auxiliary Fubini–Study area does not enter.

Theorem 2.4 extends f_∞ across infinity. Replace the root by this sphere. For each $a \in \{0, 1\}$, choose a component containing q_a , and take the path between these two components in the dual tree. Its endpoint images meet K_0 and K_1 , respectively. Omit constant components along the path; equality at the nodes preserves all intersections and endpoint incidences. At least one nonconstant component remains because $K_0 \cap K_1 = \emptyset$. Discarding the other components preserves the upper area bound and gives the required chain. $\square$

5.4. Rescaling and finite-area weighted derivatives

Lemma 5.1 is the standard affine form of Brody’s reparametrization lemma [5, Lemma 2.1]; see also [14, Section 1]. We include the proof to record the normalization and the finite-area estimate used in the recovery argument. For a single finite-area disk, a local small-area estimate proves Lemma 2.7 directly.

Lemma 5.1. *Let (X, ω) be a compact Hermitian manifold, and let $R_j > 0$. Let $f_j : U_j \rightarrow X$ be holomorphic maps, where $U_j \subset \mathbb{C}$ is open and $\overline{\mathbb{D}_{R_j}} \in U_j$, and suppose $B_{R_j}(f_j) \rightarrow \infty$. Then after affine source rescaling, a subsequence converges locally smoothly on $\mathbb{C}$ to a nonconstant entire map $u : \mathbb{C} \rightarrow X$. If in addition $\sup_j \text{Area}_{R_j}(f_j) < \infty$, then*

$$\text{Area}(u) \leq \liminf_j \text{Area}_{R_j}(f_j).$$

Proof. Under the additional area hypothesis, first pass to a subsequence for which $\text{Area}_{R_j}(f_j)$ converges to the lower limit of the original area sequence. Because f_j extends holomorphically past $\overline{\mathbb{D}_{R_j}}$, the function

$$z \mapsto (R_j - |z|)|(f_j)_z(z)|_\omega$$

extends continuously to $\overline{\mathbb{D}_{R_j}}$ by taking the value zero on the boundary. It therefore attains its maximum. For all large j this maximum is positive, so a maximizing point w_j lies in $\mathbb{D}_{R_j}$. Set $d_j := R_j - |w_j|$ and $\Lambda_j := |(f_j)_z(w_j)|_\omega$. Then

$$d_j \Lambda_j = B_{R_j}(f_j) \rightarrow \infty, \quad |(f_j)_z(z)| \leq 2 \Lambda_j \quad (|z - w_j| < d_j/2).$$

Indeed, $|z - w_j| < d_j/2$ implies $R_j - |z| > d_j/2$. Maximality then gives $(d_j/2)|(f_j)_z(z)| \leq d_j \Lambda_j$, which is the second bound. The rescaled map

$$u_j(\zeta) := f_j(w_j + \zeta/\Lambda_j), \quad |\zeta| < \frac{1}{2} d_j \Lambda_j,$$

satisfies $|(u_j)_\zeta|_\omega \leq 2$ and $|(u_j)_\zeta(0)|_\omega = 1$. The source radii $d_j \Lambda_j/2$ tend to infinity. On every fixed compact source disk, the derivative bound gives equicontinuity, and compactness of X gives a

uniformly convergent subsequence. A diagonal selection and Cauchy estimates in local target charts give a smooth limit $u : \mathbb{C} \rightarrow X$ with $|u_\zeta(0)|_\omega = 1$; hence u is nonconstant. Under the additional area hypothesis, conformal invariance gives, for each fixed $S > 0$,

$$\text{Area}_S(u) = \lim_j \text{Area}_S(u_j) \leq \lim_j \text{Area}_{R_j}(f_j) = \liminf_j \text{Area}_{R_j}(f_j).$$

Letting $S \rightarrow \infty$ proves the area bound. $\square$

Proof of Lemma 2.7. There are constants $\epsilon, c > 0$, depending only on (X, ω) , such that every holomorphic $u : \mathbb{D} \rightarrow X$ of area less than ϵ satisfies $|u_z(0)|_\omega^2 \leq c \text{Area}_1(u)$. Indeed, Lemma 1.1 of Ivashkovich and Shevchishin [28], applied with $p = 4$ and the energy convention (62), bounds $\|du\|_{L^4(\mathbb{D}_{1/2})}$ by a constant times $\text{Area}_1(u)^{1/2}$. Sobolev–Morrey, applied after a fixed smooth embedding of X into Euclidean space, then gives a small image diameter. After decreasing ϵ , the image of $\mathbb{D}_{1/2}$ lies in one uniformly controlled holomorphic chart. The submean inequality for its holomorphic coordinate derivatives gives the stated estimate.

Choose $r < R$ so that the area of h on $\{r < |w| < R\}$ is less than ϵ . For $|z| > (2r + R)/3$, the disk of radius $a = (R - |z|)/2$ about z lies in this annulus. Dilation of the local estimate gives

$$(R - |z|)^2 |h_z(z)|_\omega^2 \leq 4c \int_{\mathbb{D}_a(z)} h^* \omega \leq 4c\epsilon.$$

On the remaining compact subdisk, h_z is bounded. Thus $B_R(h) < \infty$. $\square$

6. APPLICATIONS

The relative theorem recovers the rational connectedness of Fano manifolds [6, 31]: two general points can be joined by a rational curve. Choose a positive-Ricci metric as in Proposition 3.1. For a quasifibration $q : X \dashrightarrow Y$ in Campana’s sense [6, 0.7], with $\dim Y > 0$, restrict to the dense open subsets on which it is a proper morphism with connected fibers. Generic smoothness [26, III, Corollary 10.7] makes its general fiber F smooth and compact. Theorem 1.1 gives a rational curve meeting F but not contained in it. Campana’s criterion [6, Proposition 2.7] gives rational chain connectedness. For smooth projective varieties over $\mathbb{C}$, this implies rational connectedness [2].

The pointwise construction also recovers Mori’s characterization of projective space by an ample tangent bundle [34]; for an earlier characteristic-zero proof, see Peternell [38, p. 99 and Section 2]. Let X be a connected smooth complex projective n -fold with T_X ample. The case $n = 1$ is immediate. For $n \geq 2$, the determinant $K_X^{-1} = \det T_X$ is ample [24], so X is Fano and Theorem 1.2 supplies a rational curve. Starting from this curve, two-point degree reduction gives a rational curve C with $-K_X \cdot C \leq n + 1$ [10, Section 5, Theorem 3, first step]. Let $\nu : \mathbb{P}^1 \rightarrow C \subset X$ be its normalization. Since T_X is ample,

$$\nu^* T_X \simeq \bigoplus_{i=1}^n \mathcal{O}_{\mathbb{P}^1}(a_i), \quad a_i \geq 1.$$

The nonzero differential

$$d\nu : \mathcal{O}_{\mathbb{P}^1}(2) \longrightarrow \nu^* T_X$$

forces some $a_i \geq 2$. Hence

$$-K_X \cdot C = \deg \nu^* T_X = \sum_{i=1}^n a_i \geq n + 1.$$

Thus equality holds. The resulting minimal-curve properties are those in [10, Section 5, Corollary (a),(c)]. Mori’s classification using the resulting family gives $X \simeq \mathbb{P}^n$ [34]; see also [10, Section 6]. These steps take place over $\mathbb{C}$; the analytic construction supplies the initial rational curve.

ACKNOWLEDGMENTS

We are grateful to Franc Forstnerič for introducing us to Oka geometry and for his encouragement. We also thank Jun-Muk Hwang and Mihai Păun for their helpful suggestions.

FUNDING

S.-Y. Xie was partially supported by the National Key R&D Program of China (grant nos. 2023YFA1010500 and 2021YFA1003100), the National Natural Science Foundation of China (grant nos. 12288201 and 12471081), and the Xiaomi Young Talents Program.

AI USE DISCLOSURE

The mathematical ideas and overall approach are due to the authors. AI tools assisted with exploratory work and language editing during the development of the proofs. The authors reviewed and approved every mathematical statement and take full responsibility for the paper's content.

REFERENCES

- [1] A. Alarcón and F. Forstnerič, *Oka-1 manifolds*, Math. Z. **311** (2025), Paper No. 33; doi:10.1007/s00209-025-03833-4.
- [2] C. Araujo and J. Kollár, *Rational curves on varieties*, in K. Böröczky, J. Kollár, and T. Szamuely (eds.), *Higher dimensional varieties and rational points*, Bolyai Society Mathematical Studies **12**, Springer, 2003, 13–68; doi:10.1007/978-3-662-05123-8_3.
- [3] B. Berndtsson and J.-P. Rosay, *Quasi-isometric vector bundles and bounded factorization of holomorphic matrices*, Ann. Inst. Fourier (Grenoble) **53** (2003), no. 3, 885–901; doi:10.5802/aif.1964.
- [4] R. Bhatia and T. Jain, *Higher order derivatives and perturbation bounds for determinants*, Linear Algebra Appl. **431** (2009), no. 11, 2102–2108; doi:10.1016/j.laa.2009.07.003.
- [5] R. Brody, *Compact manifolds and hyperbolicity*, Trans. Amer. Math. Soc. **235** (1978), 213–219; doi:10.1090/S0002-9947-1978-0470252-3.
- [6] F. Campana, *Connexité rationnelle des variétés de Fano*, Ann. Sci. École Norm. Sup. (4) **25** (1992), 539–545; doi:10.24033/asens.1658.
- [7] Y. Chen, J. E. Fornæss, and S.-Y. Xie, *Generating all Ahlfors currents by a single entire curve*, arXiv:2511.12473v1, 2025.
- [8] O. Debarre, *Higher-dimensional algebraic geometry*, Universitext, Springer-Verlag, 2001; doi:10.1007/978-1-4757-5406-3.
- [9] J.-P. Demailly, *Complex analytic and differential geometry*, online book, version of June 21, 2012; available online.
- [10] M. Demazure, *Caractérisations de l'espace projectif (conjectures de Hartshorne et de Frankel), d'après Shigefumi Mori*, Séminaire Bourbaki, Exp. 544 (1979/80), Lecture Notes in Math. **842**, Springer, Berlin, 1981, 11–19; published text.
- [11] F. Docquier and H. Grauert, *Levisches Problem und Rungescher Satz für Teilgebiete Steinscher Mannigfaltigkeiten*, Math. Ann. **140** (1960), 94–123; doi:10.1007/BF01360084.
- [12] Y.-H. Du, B. Guo, and S.-Y. Xie, *Analytic and algebraic Oka-1 approximation for smooth projective morphisms with rationally connected fibers*, arXiv:2609.11883v1, 2026.
- [13] J. Duval, *Sur le lemme de Brody*, Invent. Math. **173** (2008), no. 2, 305–314; doi:10.1007/s00222-008-0119-9.
- [14] J. Duval, *Around Brody lemma*, in *Hyperbolicity properties of algebraic varieties*, Panoramas et Synthèses **56** (2021), 1–12; published text.
- [15] F. Forstnerič, *Manifolds of holomorphic mappings from strongly pseudoconvex domains*, Asian J. Math. **11** (2007), 113–126; doi:10.4310/AJM.2007.v11.n1.a11.
- [16] F. Forstnerič, *The Oka principle for sections of stratified fiber bundles*, Pure Appl. Math. Q. **6** (2010), no. 3, 843–874; doi:10.4310/PAMQ.2010.v6.n3.a11.
- [17] F. Forstnerič, *Oka manifolds: From Oka to Stein and back*, with an appendix by F. Lárusson, Ann. Fac. Sci. Toulouse Math. (6) **22** (2013), no. 4, 747–809; doi:10.5802/afst.1388.
- [18] F. Forstnerič, *Stein manifolds and holomorphic mappings: The homotopy principle in complex analysis*, second edition, Ergebnisse der Mathematik und ihrer Grenzgebiete (3) **56**, Springer, Cham, 2017; doi:10.1007/978-3-319-61058-0.
- [19] F. Forstnerič, *From Stein manifolds to Oka manifolds: the h-principle in complex analysis*, in *Proceedings of the International Congress of Mathematicians 2026*, Vol. 4: Invited Lectures (Sections 5–8), SIAM, 2026, 657–677; doi:10.1137/25M1802808.

- [20] H. Grauert, *Analytische Faserungen über holomorph-vollständigen Räumen*, Math. Ann. **135** (1958), 263–273; doi:10.1007/BF01351803.
- [21] M. Gromov, *Pseudo holomorphic curves in symplectic manifolds*, Invent. Math. **82** (1985), no. 2, 307–347; doi:10.1007/BF01388806.
- [22] B. Guo and S.-Y. Xie, *Universal holomorphic maps with slow growth I: an algorithm*, Math. Ann. **389** (2024), 3349–3378; doi:10.1007/s00208-023-02719-2.
- [23] B. Guo and S.-Y. Xie, *Smooth quartic hypersurfaces of dimension at least three in projective space are Oka*, preprint, 2026; author-hosted preprint.
- [24] R. Hartshorne, *Ample vector bundles*, Publ. Math. IHÉS **29** (1966), 63–94; doi:10.1007/BF02684806.
- [25] R. Hartshorne, *Ample subvarieties of algebraic varieties*, Lecture Notes in Math. **156**, Springer-Verlag, Berlin, 1970; doi:10.1007/BFb0067839.
- [26] R. Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics **52**, Springer-Verlag, 1977; doi:10.1007/978-1-4757-3849-0.
- [27] D. T. Huynh and S.-Y. Xie, *On Ahlfors currents*, J. Math. Pures Appl. (9) **156** (2021), 307–327; doi:10.1016/j.matpur.2021.05.007.
- [28] S. Ivashkovich and V. Shevchishin, *Gromov compactness theorem for J -complex curves with boundary*, Int. Math. Res. Not. IMRN (2000), no. 22, 1167–1206; doi:10.1155/S1073792800000581.
- [29] S. Kaliman and M. Zaidenberg, *Algebraic Gromov’s ellipticity of cubic hypersurfaces*, Proc. Steklov Inst. Math. **329** (2025), 79–87; doi:10.1134/S008154382560067X.
- [30] K. Kodaira, *On Kähler varieties of restricted type*, Proc. Natl. Acad. Sci. USA **40** (1954), no. 5, 313–316; doi:10.1073/pnas.40.5.313.
- [31] J. Kollár, Y. Miyaoka, and S. Mori, *Rational connectedness and boundedness of Fano manifolds*, J. Differential Geom. **36** (1992), no. 3, 765–779; doi:10.4310/jdg/1214453188.
- [32] D. McDuff and D. Salamon, *J -holomorphic curves and quantum cohomology*, University Lecture Series **6**, American Mathematical Society, 1994; revised edition, 1995; doi:10.1090/ulect/006.
- [33] J. Milnor, *Morse theory*, Annals of Mathematics Studies **51**, Princeton University Press, Princeton, NJ, 1963.
- [34] S. Mori, *Projective manifolds with ample tangent bundles*, Ann. of Math. (2) **110** (1979), no. 3, 593–606; doi:10.2307/1971241.
- [35] S. Mori, *Cone of curves, and Fano 3-folds*, in *Proceedings of the International Congress of Mathematicians (Warszawa, 1983)*, Vol. 1, PWN, Warszawa, and North-Holland, Amsterdam, 1984, pp. 747–752; official proceedings.
- [36] S. B. Myers, *Riemannian manifolds with positive mean curvature*, Duke Math. J. **8** (1941), no. 2, 401–404; doi:10.1215/S0012-7094-41-00832-3.
- [37] A. Nicolau, *The corona property for bounded analytic functions in some Besov spaces*, Proc. Amer. Math. Soc. **110** (1990), 135–140; doi:10.1090/S0002-9939-1990-1017007-X; author-hosted full text.
- [38] T. Peternell, *Manifolds of semi-positive curvature*, in F. Catanese and C. Ciliberto (eds.), *Transcendental methods in algebraic geometry*, Lecture Notes in Math. **1646**, Springer, Berlin, 1996, 98–142; doi:10.1007/BFb0094303.
- [39] É. Picard, *Mémoire sur les fonctions entières*, Ann. Sci. École Norm. Sup. (2) **9** (1880), 145–166; doi:10.24033/asens.187.
- [40] J. Sacks and K. Uhlenbeck, *The existence of minimal immersions of 2-spheres*, Ann. of Math. (2) **113** (1981), no. 1, 1–24; doi:10.2307/1971131.
- [41] J. Simons, *Minimal varieties in Riemannian manifolds*, Ann. of Math. (2) **88** (1968), no. 1, 62–105; doi:10.2307/1970556.
- [42] Y.-T. Siu, *Dynamic multiplier ideal sheaves and the construction of rational curves in Fano manifolds*, in M. Passare (ed.), *Complex analysis and digital geometry*, Acta Universitatis Upsaliensis, Skr. Uppsala Univ. C Organ. Hist. **86**, Uppsala Universitet, 2009, 323–360.
- [43] Y.-T. Siu and S.-T. Yau, *Compact Kähler manifolds of positive bisectional curvature*, Invent. Math. **59** (1980), no. 2, 189–204; doi:10.1007/BF01390043.
- [44] G. Székelyhidi, *Greatest lower bounds on the Ricci curvature of Fano manifolds*, Compos. Math. **147** (2011), no. 1, 319–331; doi:10.1112/S0010437X10004938.
- [45] G. Tian, *On Kähler–Einstein metrics on certain Kähler manifolds with $c_1(M) > 0$* , Invent. Math. **89** (1987), no. 2, 225–246; doi:10.1007/BF01389077.
- [46] N. Wiener and P. Masani, *The prediction theory of multivariate stochastic processes. I. The regularity condition*, Acta Math. **98** (1957), 111–150.
- [47] H. Wu and S.-Y. Xie, *Entire curves generating all shapes of Nevanlinna currents*, J. Lond. Math. Soc. (2) **111** (2025), no. 5, Paper No. e70177; doi:10.1112/jlms.70177.
- [48] S.-Y. Xie, *Entire curves producing distinct Nevanlinna currents*, Int. Math. Res. Not. IMRN (2024), no. 10, 8350–8362; doi:10.1093/imrn/rnad255.
- [49] S.-Y. Xie and S. Zhao, *Periods of Oka $K3$ surfaces are dense*, arXiv:2608.24392v1, 2026.

- [50] S.-T. Yau, *On the Ricci curvature of a compact Kähler manifold and the complex Monge–Ampère equation, I*, Comm. Pure Appl. Math. **31** (1978), 339–411; doi:10.1002/cpa.3160310304.
- [51] S.-T. Yau, *Survey on partial differential equations in differential geometry*, in S.-T. Yau (ed.), *Seminar on differential geometry*, Ann. of Math. Stud. **102**, Princeton University Press, Princeton, NJ, 1982, 3–71; digital edition: doi:10.1515/9781400881918-002.